Wilson–Dirac Covariance, Majorant and Boundary Estimates

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Scope

Notes on interval restrictions of infinite-chain Wilson–Dirac occupied-band projectors: cross-boundary trace-class estimates, exact spectral majorants, reference geometry and the two-bond asymptotic, Fourier-space boundary defects, the Hilbert–Schmidt scaling limit, square-root-mass control, majorant moments and boundary corrections, and positive reference-modular traces, followed by a note that bounds the continuum cross-block interpolants on a fractional Sobolev space. Each statement keeps its stated infinite-chain, filling and scaling conditions.

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Cross-Boundary Trace-Class Estimates for Wilson–Dirac Quasifree States

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Abstract. We derive lattice Fourier moment identities, two-boundary Laplace and modified-Bessel representations, trace-class convergence and subcritical fractional Sobolev nuclear bounds for compressed infinite-chain Wilson–Dirac projectors. The finite quasifree trace-path identity and these estimates prove an entropy-difference convergence rate for this specified model. These results alone do not establish the complete Wilson relative-entropy asymptotic or its modular remainder.

Model and definitions

The model is the infinite-chain Wilson–Dirac occupied-band projection

$$P_m(k) = \frac{1}{2}\left[I - \frac{\sin k\,\sigma_x + (1+m-\cos k)\sigma_z}{\sqrt{m^2 + 4(1+m)\sin^2(k/2)}}\right], \qquad m = \mu/L.$$

The finite region consists of **L cells, each with two orbitals**. Restricting the *infinite-chain projection* to these cells is essential; the proof is not for the ground-state projection of an open finite chain. Let $U_L\colon \ell^2(\mathbb{Z};\mathbb{C}^2) \to L^2(\mathbb{R};\mathbb{C}^2)$ be the isometry whose outputs are constant on each interval $[j/L, (j+1)/L)$, and $\chi = 1_{(0,1)}$. Set

$$\begin{aligned}
\widetilde{P}_{L,\mu} &= U_L \mathsf{P}_{\mu/L} U_L^*, & X_{L,\mu} &= \chi \widetilde{P}_{L,\mu}(1-\chi),\\
P^c_\mu &= \mathcal{F}^{-1}\frac{1}{2}\left[I - \frac{u\sigma_x + \mu\sigma_z}{\sqrt{u^2+\mu^2}}\right]\mathcal{F}, & X_\mu &= \chi P^c_\mu(1-\chi),\\
Z_{L,\mu} &= X_{L,\mu} - X_{L,0}, & Z_\mu &= X_\mu - X_0,\\
\mathbb{D}_\mu &= \begin{pmatrix} 0 & Z_\mu \\ Z_\mu^* & 0 \end{pmatrix}.
\end{aligned}$$

The value of the continuum multiplier at $(u,\mu) = (0,0)$ is irrelevant. The L^2 decomposition is into the interior and **both** exterior half-lines.

Lattice Fourier moment identity (no contour deformation)

In the fixed σ_y eigenbasis, the Bloch projection has off-diagonal function

$$q_\tau(e^{ik}) = \frac{1+m-e^{ik}}{E_m(k)} = \left(1-\tau e^{ik}\right)^{1/2}\left(1-\tau e^{-ik}\right)^{-1/2}, \qquad \tau = (1+m)^{-1}.$$

For $0 < \tau < 1$ both binomial series converge absolutely and uniformly on the unit circle. Put $c_j = (-1/2)_j/j!$ and $b_j = (1/2)_j/j!$. For integer $d \geq 1$, the Fourier convention $q_d = (2\pi)^{-1}\int_{-\pi}^{\pi} e^{idk} q_\tau(e^{ik})\,dk$ yields

$$q_d(\tau) = \tau^d \sum_{j\geq 0} c_j b_{j+d} \tau^{2j} = \frac{\tau^d}{\pi} B(d+\tfrac{1}{2}, \tfrac{1}{2}) {}_2F_1(-\tfrac{1}{2}, d+\tfrac{1}{2}; d+1; \tau^2),$$

$$q_{-d}(\tau) = \tau^d \sum_{j\geq 0} c_{j+d} b_j \tau^{2j} = -\frac{\tau^d}{\pi} B(d-\tfrac{1}{2}, \tfrac{3}{2}) {}_2F_1(\tfrac{1}{2}, d-\tfrac{1}{2}; d+1; \tau^2).$$

The prefactors follow from $b_d = B(d+1/2, 1/2)/\pi$ and $c_d = -B(d-1/2, 3/2)/\pi$. Euler's beta integral, or termwise integration of its binomial expansion, therefore proves exactly

$$\boxed{\begin{aligned} q_d(\tau) &= \frac{1}{\pi} \int_0^\tau x^{d-1} \sqrt{\frac{x(1-\tau x)}{\tau - x}}\, dx, \\ q_{-d}(\tau) &= -\frac{1}{\pi} \int_0^\tau x^{d-1} \sqrt{\frac{\tau - x}{x(1-\tau x)}}\, dx. \end{aligned}} \tag{B1}$$

The integrands have integrable endpoints for $d \geq 1$; dominated convergence extends the result to $\tau = 1$, where the densities become $\sqrt{x}$ and $x^{-1/2}$. This is a separate derivation of the sign-sensitive positive- and negative-displacement formula, not an assumption about a contour.

Both boundary channels and direct continuum identification

For $\mu > 0$ set

$$a = a_L = L \log(1 + \mu/L), \quad \tau = e^{-a/L},$$

$$R_L(t) = \sqrt{\frac{\sinh((t+a)/(2L))}{\sinh((t-a)/(2L))}}, \quad R_a(t) = \sqrt{\frac{t+a}{t-a}}, \quad t > a.$$

Define $d_L^{\pm}(t) = 1_{t>a} R_L(t)^{\pm 1} - 1$ and $d_a^{\pm}(t) = 1_{t>a} R_a(t)^{\pm 1} - 1$. For $x \in (0,1),\ y > 0$, let

$$u_{L,t}(x) = e^{-t\lfloor Lx \rfloor / L}, \quad v_{L,t}(y) = e^{-t\lfloor Ly \rfloor / L}, \quad u_t(x) = e^{-tx}, \quad v_t(y) = e^{-ty}.$$

Writing $P_L(t) = |u_{L,t}\rangle\langle v_{L,t}|$ and $P(t) = |u_t\rangle\langle v_t|$, substitution $x = e^{-t/L}$ in (B1) gives **exactly**, with no asymptotic replacement,

$$H_L^+ = \int_0^\infty e^{-3t/(2L)} d_L^+(t) P_L(t)\, dt, \qquad H_L^- = \int_0^\infty e^{-t/(2L)} d_L^-(t) P_L(t)\, dt. \tag{B2}$$

The unequal exponents $3/2$ and $1/2$ result from the alternative square-root channel and the Jacobian $dx = -e^{-t/L} dt/L$. They cannot be interchanged. In the σ_y basis, the left exterior contribution to $Z_{L,\mu}$ is $(2\pi)^{-1} \left(\begin{smallmatrix} 0 & -H_L^+ \\ H_L^- & 0 \end{smallmatrix}\right)$. Reflecting the interior interval gives the right exterior contribution, with channel signs exchanged and the same $(2\pi)^{-1}$ factor. The triangle inequality over both exterior columns gives the factor $1/\pi$ in the assembled trace-norm estimate, *not* $1/(2\pi)$.

The corresponding continuum candidates are $H_b^{\pm} = \int_0^\infty d_b^{\pm}(t) P(t) dt$, with $d_b^{\pm} = 1_{t>b}[(t+b)/(t-b)]^{\pm 1/2} - 1$. These are trace-class Bochner integrals, by the estimates established below.

An additional kernel check follows. For a separation $r > 0$, substitute $t = b \cosh v$. Using $\coth(v/2) \sinh v = \cosh v + 1$ and $\tanh(v/2) \sinh v = \cosh v - 1$, the classical integral $K_j(z) = \int_0^\infty e^{-z \cosh v} \cosh(jv) dv$ yields

$$\boxed{\begin{aligned}\int_0^\infty d_b^+(t)e^{-rt}dt &= b[K_1(br)+K_0(br)]-\frac{1}{r},\\ \int_0^\infty d_b^-(t)e^{-rt}dt &= b[K_1(br)-K_0(br)]-\frac{1}{r}.\end{aligned}} \tag{B3}$$

The Fourier kernel of $q_b(u) = (b-iu)/\sqrt{u^2+b^2}$ is $b[K_0(br)+K_1(br)]/\pi$ for $r>0$, while the massless Fourier kernel is $1/(\pi r)$. For negative displacement, the K_1 term changes sign and matches the sign-reversed second channel. Equation (B3) thus identifies the continuum Laplace candidate with the **specified continuum Dirac projector** rather than merely defining a formal limiting kernel. This uses standard Bessel integral identities, without invoking a lattice branch-contour argument.

Finally, the full-line step projections converge strongly on band-limited test functions to P_b^c because the rescaled Bloch multipliers converge pointwise almost everywhere, the cell form factor tends to one, and projectors are uniformly bounded. Density and the identity $\|P_n f - Pf\|^2 = \langle f, (P_n + P - P_nP - PP_n)f\rangle$, or the standard weak-to-strong argument for projections, extend this convergence to all L^2. The trace-norm limit of the rank-one formulas then agrees with the strong limit, uniquely fixing the continuum operator and its factors.

Direct rank-one trace-norm bound and mass variation

The nuclear norms of the rank-one kernels are exactly

$$\Phi(t) := \|P(t)\|_1 = \frac{\sqrt{1-e^{-2t}}}{2t}, \qquad \|P_L(t)\|_1 = \frac{\sqrt{1-e^{-2t}}}{L(1-e^{-2t/L})}.$$

Assume $L \geq \max\{1, 2\mu\}$, so $0 < a \leq \mu \leq L/2$. On $0 < t \leq L$, the floor inequality yields

$$\|e^{-\gamma t/L}P_L(t) - P(t)\|_1 \leq 10\frac{t}{L}\Phi(t), \qquad \gamma \in \{\tfrac{1}{2}, \tfrac{3}{2}\}.$$

To control the square-root singularity at $t=a$, put $g(x) = \log(\sinh x/x)$. Since $0 \leq g'(x) \leq x/2$ for $x > 0$,

$$0 \leq \delta := \log(R_L/R_a) \leq \frac{at}{8L^2} \leq \frac{1}{16},$$

and hence

$$|R_L - R_a| + |R_L^{-1} - R_a^{-1}| \leq \frac{at}{4L^2}(R_a + R_a^{-1}).$$

The essential integrable combined densities are **exactly**

$$|d_a^+| + |d_a^-| = \begin{cases} 2, & 0 < t < a, \\ 2a/\sqrt{t^2-a^2}, & t > a. \end{cases} \tag{B4}$$

By splitting $e^{-\gamma t/L}d_L P_L - d_a P$ into its step/phase and density differences, the trace-norm bounds for the **sum of both channels** are:

Contribution	Upper bound before the two-boundary $1/\pi$ factor
Step vector and phase, $0 < t \leq L$	$10a[1+\operatorname{arcosh}(L/a)]/L$
Hyperbolic density, $a < t \leq L$	$4a/L$ (conservative; a sharper $2a/L$ follows)

Contribution	Upper bound before the two-boundary $1/\pi$ factor
Continuum tail, $t > L$	$2a/L$
Lattice tail, $t > L$	$24a/L$

For the continuum tail, use $\int_L^\infty a\,[t\sqrt{t^2-a^2}]^{-1}dt = \arcsin(a/L) \le 2a/L$. For the lattice tail, set $\epsilon = a/L \le 1/2$, $z = t/L \ge 1$, and use $\log R_L \le (\epsilon/2)\coth(z/4) \le 5\epsilon/2$, $R_L - 1 \le 5\epsilon$, $\|P_L\|_1 \le 6/(5L)$, and the exponentially decaying phase factors. These reproduce the stated bound without numerical extrapolation.

Define $\mathcal{Z}_b := Z_b$ for the continuum cross-boundary difference at mass b. Consequently

$$\boxed{\|Z_{L,\mu} - \mathcal{Z}_a\|_1 \le \frac{a}{\pi L}\left[40 + 10\log\frac{2L}{a}\right].} \tag{B5}$$

To change the effective mass a to μ, rewrite the continuum channels as $H_b^\pm = \int_0^\infty b\,d_\pm(v)P(bv)\,dv$, where $d_\pm(v) = 1_{v>1}[(v+1)/(v-1)]^{\pm 1/2} - 1$. The derivative of $bP(bv)$ is a sum of three rank-one kernels, and elementary truncated-exponential moment bounds give

$$\left\|\frac{d}{db}[bP(bv)]\right\|_1 \le (1+\sqrt{2})\Phi(bv).$$

The combined density $w(v) = |d_+(v)| + |d_-(v)|$ is 2 below 1 and $2/\sqrt{v^2-1}$ above it. With $\Phi(t) \le 1/\sqrt{2t}$,

$$\frac{1}{\pi}\int_0^\infty w(v)\Phi(bv)dv \le \frac{C_{\text{ent}}}{2\sqrt{b}}, \quad C_{\text{ent}} = \frac{2\sqrt{2}}{\pi}\left(2 + \frac{\sqrt{\pi}\,\Gamma(1/4)}{2\Gamma(3/4)}\right).$$

Dominated differentiation on compact positive mass intervals is valid (the integrand is $O(v^{-1/2})$ at zero, $O((v-1)^{-1/2})$ at one, and $O(v^{-3/2})$ at infinity). Thus

$$\|\mathcal{Z}_b'\|_1 \le \frac{(1+\sqrt{2})C_{\text{ent}}}{2\sqrt{b}}, \qquad \|\mathcal{Z}_b - \mathcal{Z}_c\|_1 \le (1+\sqrt{2})C_{\text{ent}}|\sqrt{b} - \sqrt{c}|.$$

Since $0 \le \mu - a_L \le \mu^2/(2L)$, the calculation gives the claimed finite-size bound:

$$\boxed{\begin{aligned}&\|Z_{L,\mu} - Z_\mu\|_1 \le B_{L,\mu},\\ &\qquad B_{L,\mu} = \frac{a_L}{\pi L}\left[40 + 10\log\frac{2L}{a_L}\right] + (1+\sqrt{2})C_{\text{ent}}(\sqrt{\mu} - \sqrt{a_L}).\end{aligned}} \tag{B6}$$

For fixed $\mu > 0$, this is $O_\mu((\log L)/L)$. For each $M > 0$, a common bound for $0 \le \mu \le M$, $L \ge 2M$, follows from monotonicity of $a[40 + 10\log(2L/a)]$ on $(0, M]$:

$$\sup_{0\le\mu\le M}\|Z_{L,\mu} - Z_\mu\|_1 \le \frac{M}{\pi L}\left[40 + 10\log\frac{2L}{M}\right] + \frac{(1+\sqrt{2})C_{\text{ent}}M^{3/2}}{2L}.$$

At $\mu = 0$, both differences are precisely zero, not an unexamined singular endpoint.

direct $0 < s < 1/2$ Sobolev nuclear estimate

For $0 < s < 1/2$, characteristic multiplication by either half-line, hence by χ and $1-\chi$, is bounded on $H^s(\mathbb{R})$. One direct proof starts with the Gagliardo seminorm. The extra cutoff crossing term is bounded by $C_s \int |f(x)|^2 |x|^{-2s} dx$. Under Fourier transformation its bilinear positive kernel is proportional to

$$K_s(u,v) = \frac{|u-v|^{2s-1}}{|u|^s |v|^s}.$$

The weighted Schur test with weight $|u|^{-1/2}$ reduces precisely to $\int_{\mathbb{R}} |1-t|^{2s-1} |t|^{-s-1/2} dt < \infty$. The exponent conditions at $0, 1, \infty$ give $0 < s < 1/2$. There is no assertion at $s = 1/2$, where nonzero boundary traces already obstruct sharp cutoffs.

For the continuum rank-one vectors, the Fourier transforms of the zero extensions are

$$\hat{u}_t(u) \propto \frac{1 - e^{-t-iu}}{t+iu}, \qquad \hat{v}_t(u) \propto \frac{1}{t+iu}.$$

Direct integration after splitting $t \le 1$ and $t \ge 1$ gives

$$\begin{aligned} &\|v_t\|_{H^s} \le C_s (t^{-1/2} + t^{s-1/2}), \\ &\|u_t\|_{H^s} \le C_s \quad (t \le 1), \qquad \|u_t\|_{H^s} \le C_s t^{s-1/2} \quad (t \ge 1). \end{aligned}$$

Together with the elementary L^2 norms, both rank-one nuclear output norms $\|u_t\|_{H^s} \|v_t\|_2$ and $\|v_t\|_{H^s} \|u_t\|_2$ are bounded by $C_s t^{-1/2}$ **at all positive** t. The inequality $s - 1 \le -1/2$ for $t \ge 1$ is the relevant global step. Equation (B4) gives the exact scalar integral

$$\int_0^\infty (|d_\mu^+| + |d_\mu^-|) t^{-1/2} dt = \sqrt{\mu} \left[4 + \frac{\sqrt{\pi}\,\Gamma(1/4)}{\Gamma(3/4)} \right].$$

Since the left and right exterior channels, their adjoints, and the fixed finite spin matrices are all included, the rank-one Bochner integrals converge in the operator nuclear ideal $\mathcal{S}_1(L^2, H^s)$ and prove

$$\boxed{\|\mathbb{D}_\mu\|_{\mathcal{S}_1(L^2,H^s)} \le C_s \sqrt{\mu}, \qquad 0 < s < 1/2, \quad \mu \ge 0.} \tag{B7}$$

This statement is about a nuclear operator **from** L^2 **into** H^s, not the eigenvalues of a single unbounded Sobolev multiplier; no commutation is required.

Finite cross-block trace path

Lemma (Finite cross-block trace path). Let P be an orthogonal projection, let χ have finite rank, and put $C = \chi P \chi$ and $X = \chi P (1-\chi)$. With

$$\mathbb{H}(X) = \begin{pmatrix} 0 & X \\ X^* & 0 \end{pmatrix}, \qquad F(t) = h\left(\frac{1+\sqrt{1-4t^2}}{2} \right), \quad h(p) = -p \log p - (1-p)\log(1-p),$$

one has $XX^* = C - C^2$, $\|\mathbb{H}(X)\| \le 1/2$, and $S(C) = \frac{1}{2} \operatorname{Tr} F(\mathbb{H}(X))$. For the embedded finite cross blocks, define

$$\mathbb{D}_{L,\mu} = \mathbb{H}(Z_{L,\mu}), \quad A_{L,\theta} = \mathbb{H}(X_{L,0}) + \theta \mathbb{D}_{L,\mu}, \quad Y_{L,\theta} = A_{L,\theta} + I/2,$$

and, in the continuum,

$$A_\theta = \mathbb{H}(X_0) + \theta \mathbb{D}_\mu, \qquad Y_\theta = A_\theta + I/2.$$

The interpolants $Y_{L,\theta}, Y_\theta$ are positive contractions and

$$\Delta S_L(\mu) = \frac{1}{2}\int_0^1 \mathrm{Tr}[F'(A_{L,\theta})\mathbb{D}_{L,\mu}]\,d\theta.$$

The analogous continuum integral $\mathcal{S}(\mu) = \frac{1}{2}\int_0^1 \mathrm{Tr}[F'(A_\theta)\mathbb{D}_\mu]d\theta$ is finite whenever $\mathbb{D}_\mu$ is trace class.

Proof. The (χ,χ) block of $P^2 = P$ gives $XX^* = C - C^2 \le I/4$. A singular-value decomposition of X gives signed eigenvalue pairs $\pm s_j$ of $\mathbb{H}(X)$. Since $F(\sqrt{p(1-p)}) = h(p)$ and $F(0) = 0$, the entropy identity follows. For the embedded finite operators the interval is infinite dimensional as an L^2 subspace, but $X_{L,\mu}$ has finite rank and the compression is zero off its finite step-function subspace, so the same argument applies. The norm bound $1/2$ is preserved by convex interpolation, proving the contraction claims. The ranges of the two finite endpoint block operators span a common finite-dimensional invariant subspace. On it F is C^1 on $[-1/2, 1/2]$, with $F'(0) = 0$, and the trace derivative is $\mathrm{Tr}[F'(A_{L,\theta})\mathbb{D}_{L,\mu}]$. This identity follows, for example, by uniformly approximating F' by polynomials and applying cyclicity to their primitives before passing to the limit in finite dimension. Integration proves the path formula. Finally $\|F'\|_\infty \le 2$, so the continuum product is trace class and its trace is continuous in θ by continuous functional calculus. The continuum integral is absolutely finite; no difference of divergent continuum entropies is taken. □

Uniform projector comparison and entropy consequence

Let $p_{\mu/L}(k)$ be the lattice Bloch projection, and $p_\mu(u)$ the continuum multiplier. On $|u| \le \pi L$, write

$$q_L(u) = \mathrm{sinc}^2(u/(2L)), \quad a = (L\sin(u/L), \mu + L[1-\cos(u/L)]), \quad b = (u,\mu).$$

The Pauli projector formula, $\|a/|a| - b/|b|\| \le 2|a-b|/|b|$, and elementary Taylor-remainder inequalities give $\|p_{\mu/L}(u/L) - p_\mu(u)\| \le |u|/(\sqrt{2}L)$, uniformly in $\mu \ge 0$. **The output can contain aliases.** Rather than dropping them, use that both full-line operators are orthogonal projections to derive the exact band-limited squared-norm identity

$$\|[\widetilde{P}_{L,\mu} - P^c_\mu]f\|_2^2 = \int q_L \|[p_{\mu/L}(u/L) - p_\mu(u)]\hat{f}(u)\|^2 du + \int (1-q_L)\|p_\mu(u)\hat{f}(u)\|^2 du.$$

Since $1 - q_L \le u^2/(12L^2)$ on the band and the difference of orthogonal projections has norm at most one off it, frequency splitting proves

$$\boxed{\|\widetilde{P}_{L,\mu} - P^c_\mu\|_{H^s \to L^2} \le C_s L^{-s}} \qquad (0 < s < 1/2), \tag{B8}$$

with C_s independent of L, μ. The cutoff multiplier lemma supplies the same estimate for the self-adjoint off-diagonal interpolants, and their continuum versions act boundedly on H^s.

For the entropy branch, set $F(t) = h((1+\sqrt{1-4t^2})/2)$, $h(p) = -p\log p - (1-p)\log(1-p)$, and $g(z) = F'(z/2)$. The exact identity

$$g(z) = -2z\log|z| + r(z), \quad r \in C^2([-1,1]),$$

implies that the cosine–Chebyshev coefficients of g are $O(k^{-2})$. For the logarithmic term the odd coefficients at $k = 2j+1 \ge 3$ are exactly $(-1)^j/[j(j+1)]$, with even coefficients zero. Thus the degree-n Chebyshev truncation approximates g uniformly within C/n; when written as a polynomial in $Y \in [0,1]$, its monomial coefficient ℓ^1-norm is at most $C7^n$.

For each θ, abbreviate $Y_L = Y_{L,\theta}$ and $Y = Y_\theta$ from the trace-path lemma. The noncommutative telescoping identity

$$(Y_L^k - Y^k)\mathbb{D}_\mu = \sum_{j=0}^{k-1} Y_L^j (Y_L - Y) Y^{k-1-j} \mathbb{D}_\mu$$

places the **only** high-regularity input to the right of the $H^s \to L^2$ difference. Left lattice factors are merely L^2 contractions, so no unjustified lattice H^s functional calculus is used. Taking the nuclear norm and applying (B7)–(B8) yields

$$|\Delta S_L(\mu) - \mathcal{S}(\mu)| \le 2B_{L,\mu} + \frac{C}{n}\|\mathbb{D}_\mu\|_1 + C_s n (7M_s)^n L^{-s} \|\mathbb{D}_\mu\|_{\mathcal{S}_1(L^2,H^s)}.$$

Choosing $s = 1/4$ and $n = \lfloor s \log L / [2\log(7M_s)] \rfloor$ (for sufficiently large L) establishes

$$\boxed{\sup_{0\le\mu\le M} |\Delta S_L(\mu) - \mathcal{S}(\mu)| = O_M(1/\log L)}. \tag{B9}$$

Here $\mathcal{S}(\mu)$ is the finite continuum **trace path**, not the difference of two separately divergent continuum entropies. The finite trace-path lemma above proves the spectral-calculus and singular-value identities used in this assertion. Thus (B9) is a theorem for the specified boundary subsystem; it does not assume a full relative-entropy finite-part limit.

Scope and limitations

The estimates concern restrictions of infinite-chain quasifree projections, not the occupied projector of an open finite chain. The derived boundary estimates and entropy-difference rate do not by themselves imply modular or nonlinear convergence. Those separate inputs are proved for this model in *Finite Part of Wilson–Dirac Relative Entropy*, whose Theorem 1 assembles the full fixed-μ expansion, and are sharpened in *Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals*. No uniform Coulomb or moving-edge matching estimate is asserted as a consequence of the boundary argument.

Mathematical references

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Exact Spectral Majorants for a Wilson-Dirac Interval

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Abstract

For the negative-energy band of an infinite one-dimensional Wilson–Dirac chain, compressed to a finite interval of consecutive two-orbital cells, we construct an exact longitudinal–transverse decomposition of the difference between the massive and massless covariances. A positive Toeplitz majorant controls the longitudinal component, while the transverse component satisfies a pairing inequality in the eigenbasis of the massless reference. Direct Fourier analysis and finite Jacobi bond identities establish, at every interval length, a critical inverse-linear spectral-tail bound with explicit constants. The resulting estimates apply to the fixed-parameter mass scaling of the relative-entropy problem and establish the positive-majorant and staggered-pairing conditions required in its modular remainder analysis. We also obtain an exact finite-interval Hilbert–Schmidt majorization. The reference-modular approximation and the other asymptotic branches require separate analysis.

1. Covariance and reference geometry

For a dimensionless mass $m \geq 0$, let

$$h_m(k) = \sin k\, \sigma_x + (1 + m - \cos k)\sigma_z, \qquad P_m(k) = \frac{1}{2}\left(I - \frac{h_m(k)}{E_m(k)}\right),$$

$$E_m(k)^2 = m^2 + 4(1+m)\sin^2(k/2), \qquad -\pi < k < \pi.$$

The multiplier P_m is an orthogonal projection for almost every momentum. Its compression to L consecutive cells defines the $2L \times 2L$ covariance $C_{L,m}$. We use $N = 2L$, fix $\mu > 0$, and set

$$m = \frac{\mu}{L}, \qquad r = \frac{m}{m+2} = \frac{\mu}{N+\mu} \in (0,1).$$

The on-site unitaries

$$W = \frac{I + i\sigma_x}{\sqrt{2}}, \qquad D = \mathrm{diag}(-i, 1)$$

followed by cell staggering $(-1)^a$ give an interval-preserving unitary U_L. Fourier integration of the massless symbol gives

$$B_N := U_L C_{L,0} U_L^*, \qquad (B_N)_{ij} = \begin{cases} \frac{1}{2}, & i = j, \\ \dfrac{\sin[\pi(i-j)/2]}{\pi(i-j)}, & i \neq j. \end{cases}$$

Indeed, before staggering, the rotated massless projector has upper-right Bloch entry $i\,\mathrm{sgn}(k)e^{-ik/2}/2$. In the Fourier convention above, the elementary integral

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} \mathrm{sgn}(k) e^{i(n+1/2)k} dk = \frac{i}{\pi(n+1/2)} \qquad (n \in \mathbb{Z})$$

and its companion with $n - 1/2$ give the displayed coefficients after the diagonal phase transformation and staggering. The interval compression of the half-band Fourier projector obeys $0 < B_N < I$.

Define the real Jacobi matrix J_N by its off-diagonal entries

$$(J_N)_{j-1,j} = (J_N)_{j,j-1} = \alpha_j := \frac{j(N-j)}{N}, \qquad 1 \le j < N,$$

with $\alpha_0 = \alpha_N = 0$. An entrywise calculation, using $(d-1)T_{d-1} + (d+1)T_{d+1} = 0$ for $T_d = \sin(\pi d/2)/(\pi d)$, establishes $[B_N, J_N] = 0$. The Jacobi matrix is irreducible, so its spectrum is simple. Hence the two reference matrices admit a common orthonormal eigenbasis.

Throughout, $A_N = U_L C_{L,m} U_L^*$ and $\Delta_N = A_N - B_N$. These are *compressions of the full-line band projections*, not finite-chain open-boundary ground-state projectors.

2. Exact longitudinal–transverse decomposition

For $k \in (-\pi, 0) \cup (0, \pi)$, write

$$x = \sin(k/2), \quad c = \cos(k/2), \quad d_r = \sqrt{x^2 + r^2 c^2}, \quad u_r = \frac{|x|}{d_r}, \quad a_r = \frac{rc}{d_r}, \quad g_r = 1 - u_r.$$

Notice $0 \le g_r \le 1$ and $u_r^2 + a_r^2 = 1$. In the on-site DW-rotated Bloch basis, the two Hermitian involutions

$$Q_0(k) = \begin{pmatrix} 0 & i\,\mathrm{sgn}(k) e^{-ik/2} \\ -i\,\mathrm{sgn}(k) e^{ik/2} & 0 \end{pmatrix}, \qquad Q_\perp(k) = \begin{pmatrix} 0 & e^{-ik/2} \\ e^{ik/2} & 0 \end{pmatrix}$$

satisfy

$$Q_0^2 = Q_\perp^2 = I, \qquad Q_0 Q_\perp + Q_\perp Q_0 = 0.$$

In this basis the occupied-band involution is $Q_m(k) := 2(DW)P_m(k)(DW)^* - I = u_r Q_0 + a_r Q_\perp$. Consequently the difference of full-line projectors has the exact decomposition

$$P_m - P_0 = \mathsf{S} + \mathsf{D}, \qquad \mathsf{S}(k) = -\frac{g_r(k)}{2} Q_0(k), \quad \mathsf{D}(k) = \frac{a_r(k)}{2} Q_\perp(k), \quad \mathsf{G}(k) = \frac{g_r(k)}{2} I. \tag{2.1}$$

Here the full-line objects are understood after the on-site rotation; the cell-staggering conjugation can be applied to all of them simultaneously. Fibrewise,

$$-\mathsf{G} \le \mathsf{S} \le \mathsf{G}, \quad [\mathsf{G}, Q_0] = 0, \quad \{\mathsf{D}, Q_0\} = 0, \tag{2.2}$$

$$\mathsf{D}^2 = \mathsf{G} - \mathsf{G}^2 \le \mathsf{G}, \qquad \boxed{(P_m - P_0)^2 = \mathsf{G}}. \tag{2.3}$$

All identities hold as full-line Fourier multipliers; exceptional momentum points of measure zero do not affect the operators. Denote the interval compressions, including staggering, by S_N, D_N, G_N, respectively. Then $\Delta_N = S_N + D_N$ exactly.

Theorem 1 (positive majorant and exact trace). For every $L \ge 1$ and $\mu > 0$,

$$\boxed{-G_N \le S_N \le G_N, \qquad G_N \ge 0, \qquad \mathrm{Tr}\, G_N = \frac{L}{\pi}\left(\pi - \frac{2\arccos r}{\sqrt{1-r^2}}\right) \le \frac{\mu}{\pi}.} \tag{2.4}$$

Proof. Compression preserves Loewner order in (2.2). Since G is a scalar two-orbital Fourier multiplier,

$$\operatorname{Tr} G_N = \frac{L}{2\pi}\int_{-\pi}^{\pi} g_r(k)\,dk = \frac{L}{\pi}\int_0^{\pi} g_r(k)\,dk.$$

With $q = k/2$, substitution $z = \cos q$ gives

$$\int_0^{\pi} g_r(k)\,dk = \pi - 2\int_0^1 \frac{dz}{\sqrt{1-(1-r^2)z^2}} = \pi - \frac{2\arccos r}{\sqrt{1-r^2}}.$$

Let $M(r)$ denote this last expression, with $M(0) = 0$. Its derivative is

$$M'(r) = \frac{2}{1-r^2} - \frac{2r\arccos r}{(1-r^2)^{3/2}} \le 2,$$

because $\arccos r \ge r\sqrt{1-r^2}$, equivalently $\theta \ge \sin\theta\cos\theta$ for $\theta = \arccos r \in [0,\pi/2]$. Thus $M(r) \le 2r$, and $2Lr = \mu N/(N+\mu) \le \mu$. □

Theorem 2 (staggered pairing). Let v be any normalized vector supported in the selected interval. Set $b = \langle v, B_N v\rangle$, $q = \langle v, G_N v\rangle$. Then

$$\boxed{|\langle v, D_N v\rangle| \le 2\sqrt{q\,b(1-b)}.} \tag{2.5}$$

In particular this holds for every common normalized reference eigenvector of B_N and J_N.

Proof. Work before compression with the full-line massless projector P_0, transverse operator D, and commuting positive operator G. Equation (2.2) implies $P_0\mathsf{D}P_0 = (I-P_0)\mathsf{D}(I-P_0) = 0$. Write $p = P_0 v$, $z = (I-P_0)v$, and let

$$X = \langle p, \mathsf{G}p\rangle, \qquad Y = \langle z, \mathsf{G}z\rangle.$$

Because $[\mathsf{G}, P_0] = 0$, we have $X + Y = q$, while $\|p\|^2 = b$ and $\|z\|^2 = 1-b$. Moreover,

$$|\langle v, \mathsf{D}v\rangle| = 2|\operatorname{Re}\langle p, \mathsf{D}z\rangle| \le 2\min\{\sqrt{X(1-b)}, \sqrt{Yb}\},$$

by Cauchy–Schwarz and $\mathsf{D}^2 \le \mathsf{G}$. For nonnegative X, Y with $X + Y = q$, $\min\{X(1-b), Yb\} \le b(1-b)q$: if $X \le bq$, use the first term; otherwise $Y < (1-b)q$, and use the second. Since v is interval-supported, its full-line expectation equals $\langle v, D_N v\rangle$. □

3. A uniform critical Jacobi spectral tail

Let (λ_j, v_j) be the eigenpairs of J_N, with $\|v_j\| = 1$, and define the positive measure

$$\nu_N := \sum_{j=1}^{N} \langle v_j, G_N v_j\rangle\, \delta_{|\lambda_j|}.$$

Theorem 3 (explicit critical tail). For every even $N = 2L \ge 2$, every $\mu > 0$, and every $t > 0$,

$$\boxed{\nu_N((t,\infty)) \le \min\left\{\frac{\mu}{\pi}, \sqrt{\frac{2}{15}}\frac{\mu^2}{\pi t} + \frac{\mu}{3\pi t^2}\right\}.} \tag{3.1}$$

Proof. The first estimate is Theorem 1. To obtain the second, express the same-parity blocks of G_N as Fourier mixtures. For $\varepsilon \in \{0,1\}$ define the length-N vector

$$\phi_{k,\varepsilon}(2a+\varepsilon) = (-1)^a e^{ika}, \quad \phi_{k,\varepsilon}(2a+1-\varepsilon) = 0, \qquad 0 \le a < L.$$

The positive Toeplitz representation is

$$G_N = \frac{1}{2}\sum_{\varepsilon=0}^{1} \int_{-\pi}^{\pi} g_r(k) |\phi_{k,\varepsilon}\rangle\langle\phi_{k,\varepsilon}| \, \frac{dk}{2\pi}, \qquad \|\phi_{k,\varepsilon}\|^2 = L. \tag{3.2}$$

For either parity, application of J_N yields adjacent-bond differences of the form $\alpha_j - e^{ik}\alpha_{j+1}$, up to phases. Reflection about the centre of the interval interchanges the two parities, making their squared-norm sums identical. Direct polynomial summation gives

$$\frac{1}{L}\sum_{j \equiv \varepsilon\,(2)} (\alpha_j - \alpha_{j+1})^2 = \frac{N^2-1}{3N^2},$$

$$\frac{4}{L}\sum_{j \equiv \varepsilon\,(2)} \alpha_j \alpha_{j+1} = \frac{2(N^2-1)(N^2-4)}{15N^2},$$

where the sums run over $0 \le j < N$, and $\alpha_0 = \alpha_N = 0$. Consequently

$$\frac{\|J_N \phi_{k,\varepsilon}\|^2}{L} = \frac{N^2-1}{3N^2} + \frac{2(N^2-1)(N^2-4)}{15N^2}\sin^2(k/2) \le \frac{1}{3} + \frac{N^2 k^2}{30}. \tag{3.3}$$

For $\Pi_t = \mathbf{1}_{\{|J_N| > t\}}$, the spectral theorem implies

$$\|\Pi_t \phi\|^2 \le \min\{\|\phi\|^2, t^{-2}\|J_N\phi\|^2\}.$$

Substituting (3.3) into (3.2) and using the evenness of g_r,

$$\nu_N((t,\infty)) \le \frac{L}{\pi}\int_0^{\pi} g_r(k) \min\left\{1, \frac{1}{3t^2} + \frac{N^2k^2}{30t^2}\right\} dk. \tag{3.4}$$

The elementary scalar bounds

$$\int_0^{\pi} g_r(k)\,dk \le 2r, \qquad g_r(k) = 1 - \left[1 + r^2\cot^2(k/2)\right]^{-1/2} \le \frac{2r^2}{k^2} \quad (0 < k \le \pi) \tag{3.5}$$

follow from Theorem 1, $1-(1+x)^{-1/2} \le x/2$, and $\tan(k/2) \ge k/2$. Set $k_* := \sqrt{30}\,t/N$. Bound the integrand in (3.4) by the second argument on $[0, \min(k_*, \pi)]$, and by 1 on $[k_*, \pi]$ when the latter interval is nonempty. Equation (3.5) yields, whether or not $k_* > \pi$,

$$\begin{aligned}\nu_N((t,\infty)) &\le \frac{L}{\pi}\left[\frac{2r}{3t^2} + \frac{N^2}{30t^2}(2r^2 k_*) + \frac{2r^2}{k_*}\right] \\ &= \frac{L}{\pi}\left[\frac{2r}{3t^2} + \frac{4Nr^2}{\sqrt{30}\,t}\right].\end{aligned}$$

Finally $2Lr = Nr \le \mu$ and $4LNr^2 = 2N^2r^2 \le 2\mu^2$. Since $2/\sqrt{30} = \sqrt{2/15}$, this is exactly the second estimate in (3.1). □

4. A finite compression inequality and continuum consequence

Corollary 4 (covariance-distance domination). At every L and $\mu > 0$,

$$\boxed{0 \le \Delta_N^2 \le G_N, \qquad \|A_N - B_N\|_{\mathrm{HS}}^2 \le \operatorname{Tr} G_N \le \frac{\mu}{\pi}.} \tag{4.1}$$

Proof. If χ is the interval projection and $T = P_m - P_0$ after the same full-line unitary, then $T = T^*$ and $T^2 = \mathsf{G}$ by (2.3). The identity

$$\chi T^2 \chi - (\chi T \chi)^2 = \chi T (I - \chi) T \chi \ge 0$$

proves the operator inequality; take the finite trace and use Theorem 1. □

For μ fixed and $L \to \infty$, the exact trace in (2.4) tends to μ/π, since g_r has integrated mass $2r + O(r^2)$. Thus whenever the finite covariance differences converge in Hilbert–Schmidt norm to the continuum kernel of *Hilbert-Schmidt Scaling Limit for Wilson-Dirac Covariances*, with squared norm $H(\mu)$, (4.1) independently yields

$$0 \le H(\mu) \le \frac{\mu}{\pi}.$$

There is an independent continuum verification. For the rescaled Bloch-projector difference

$$F_\mu(t) = -\frac{1}{2}\left[\left(\frac{t}{\sqrt{t^2+\mu^2}} - \operatorname{sgn} t\right)\sigma_x + \frac{\mu}{\sqrt{t^2+\mu^2}}\sigma_z\right],$$

one calculates $\|F_\mu(t)\|_F^2 = 1 - |t|/\sqrt{t^2+\mu^2}$ and therefore, by direct integration and Plancherel,

$$\int_{\mathbb{R}} \|\widehat{F}_\mu(s)\|_F^2 ds = \frac{1}{2\pi}\int_{\mathbb{R}} \|F_\mu(t)\|_F^2 dt = \frac{\mu}{\pi}.$$

The interval overlap formula for $H(\mu)$ has weight $0 \le (1 - |s|)_+ \le 1$, so the same bound follows without reference to finite-size eigenvalues.

The finite identity also identifies the trace deficit as cross-interval leakage:

$$\operatorname{Tr} G_N - \|\Delta_N\|_{\mathrm{HS}}^2 = \|(I - \chi)T\chi\|_{\mathrm{HS}}^2.$$

5. Consequences for the relative-entropy problem

Theorems 1–3 establish for the specified Wilson chain the Fourier splitting, positive-majorant trace bound, transverse reference-mode pairing, and critical Jacobi spectral tail. The proofs require no large-L hypothesis and remain valid for arbitrary positive μ at each finite L.

In the relative-entropy decomposition, the modular term involves $E_N = \log(I - B_N) - \log B_N + N\eta(J_N/N)$. The "Positive weighted modular control" theorem in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*, section "Global envelope and weighted consequences," supplies a bounded longitudinal term and the variance estimate $O((\log N)/N^2)$. The entropy trace-path limit is established in *Cross-Boundary Trace-Class Estimates for Wilson–Dirac Quasifree States*, equation (B9). For the specified compressed infinite-chain Wilson model at fixed $\mu > 0$, the additional convergence estimates and full constant-term expansion are established in *Finite Part of Wilson–Dirac Relative Entropy*, Theorem 1 and its proof, and sharpened in *Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals*, Theorem 3, Corollary 4 and Theorem 6. Their exact concentration-connection and boundary inputs remain those stated in those papers. The present paper supplies the projector, pairing, and positive-tail inputs.

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Exact Reference Geometry and the Two-Bond Asymptotic of a Wilson–Dirac Covariance

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Abstract. For the specified one-dimensional two-orbital Wilson–Dirac Bloch Hamiltonian, an explicit cellwise, interval-preserving unitary transforms every finite compression of the massless occupied-band projection into the scalar half-filled sine-kernel matrix. The resulting discrete-prolate sine kernel commutes exactly with a tridiagonal Jacobi matrix of simple spectrum. The same unitary gives an exact two-bond reduction of the massive-minus-massless Jacobi trace, its complete-elliptic-integral representation, and its fixed-positive-mass logarithmic expansion. The proofs work for every positive interval length and do not use numerical extrapolation. These identities do not resolve the distinct nonlinear, Coulomb/reference-mode, or entropy estimates needed for a full relative-entropy asymptotic theorem.

1. Model and notation

Let $\sigma_x, \sigma_y, \sigma_z$ denote the Pauli matrices and, for $m \geq 0$, put

$$h_m(k) = \sin k\, \sigma_x + (m + 1 - \cos k)\, \sigma_z, \qquad E_m(k) = \left[m^2 + 4(m+1)\sin^2(k/2)\right]^{1/2}.$$

The infinite-chain negative-band spectral projection has symbol $P_m(k) = \frac{1}{2}(I - h_m(k)/E_m(k))$, interpreted almost everywhere for $m = 0$. Let

$$(C_{L,m})_{ab} = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{ik(a-b)} P_m(k)\, dk, \qquad 0 \leq a, b < L,$$

where each entry is a two-by-two orbital block. Set $N = 2L$. We distinguish this compression of an infinite-volume spectral projection from the negative spectral projection of an open finite-chain Hamiltonian.

2. Explicit all-interval scalarization

Theorem 1 (exact massless reference scalarization). Define

$$H_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad W = \frac{I - i\sigma_x}{\sqrt{2}} H_2, \qquad R = \sigma_z,$$

and let $U = \bigoplus_{j=0}^{L-1} (-1)^j RW$. Then, for every integer $L \geq 1$,

$$\boxed{UC_{L,0}U^\dagger = B_N}, \qquad (B_N)_{pq} = \begin{cases} \frac{1}{2}, & p = q, \\ \dfrac{\sin(\pi(p-q)/2)}{\pi(p-q)}, & p \neq q, \end{cases}$$

with $0 \leq p, q < N$. The unitary is a direct sum over cells, is independent of the interval length except for repetition of its fixed blocks, and does not mix sites inside and outside the interval.

Proof. For an integer cell displacement d, direct Fourier integration at $m = 0$ gives

$$(C_{L,0})_{ab} = \tfrac{1}{2}\delta_{d0} I + x_d \sigma_x + z_d \sigma_z, \qquad d = a - b,$$

where

$$x_d = -\frac{i}{2\pi} \frac{d}{d^2 - 1/4}, \qquad z_d = -\frac{1}{\pi(1 - 4d^2)}.$$

For clarity, these follow from $\sin k / E_0(k) = \operatorname{sgn}(k) \cos(k/2)$ and $(1 - \cos k)/E_0(k) = |\sin(k/2)|$, using elementary sine–cosine product integrals on $[0, \pi]$. In particular $x_0 = 0$, $x_{-d} = \overline{x_d}$, and $z_{-d} = z_d$.

Direct multiplication of the constant two-by-two matrices establishes $W \sigma_z W^\dagger = \sigma_x$ and $W \sigma_x W^\dagger = -\sigma_y$. Conjugating by $R = \sigma_z$ therefore produces

$$U_a (C_{L,0})_{ab} U_b^\dagger = (-1)^d \left(\tfrac{1}{2}\delta_{d0} I + x_d \sigma_y - z_d \sigma_x \right).$$

The upper-right and lower-left entries are respectively

$$(-1)^d (-z_d - i x_d) = \frac{(-1)^{d+1}}{\pi(2d - 1)}, \qquad (-1)^d (-z_d + i x_d) = \frac{(-1)^d}{\pi(2d + 1)}.$$

These are exactly the sine-kernel entries at scalar index differences $2d - 1$ and $2d + 1$. The like-orbital entries are zero for $d \neq 0$ and equal to $1/2$ for $d = 0$. Thus every matrix entry agrees, proving the asserted identity for all L. □

Corollary 2 (reference positivity, commuting Jacobi operator, and simple spectrum). Define

$$(J_N)_{j-1,j} = (J_N)_{j,j-1} = \frac{j(N - j)}{N}, \quad 1 \le j < N,$$

with all remaining entries zero. Then $0 < B_N < I$, $[B_N, J_N] = 0$, and J_N has simple real spectrum.

Proof. The sine kernel is the compression of the Fourier projection onto the momentum arc $[-\pi/2, \pi/2]$. For any nonzero vector $v \in \mathbb{C}^N$, its associated trigonometric polynomial is nonzero on subsets of positive measure in both this arc and its complement, because a nonzero trigonometric polynomial cannot vanish on an open arc. Hence $0 < \langle v, B_N v \rangle < \|v\|^2$.

Put $b_d = \sin(\pi d/2)/(\pi d)$ with $b_0 = 1/2$, and $a_j = j(N - j)/N$, with $a_0 = a_N = 0$. Direct evaluation gives

$$[J_N, B_N]_{pq} = \frac{N - p - q - 1}{N} \left[(p - q - 1) b_{p-q-1} + (p - q + 1) b_{p-q+1} \right] = 0.$$

The last equality follows from the sine addition formula, including the removable zero-index terms. All $a_j > 0$ for $1 \le j < N$. For an eigenvector of the symmetric Jacobi matrix, its first coordinate fixes the others recursively; a zero first coordinate forces the zero vector. Every eigenspace is therefore one-dimensional, and the real symmetric matrix has simple spectrum. □

3. The exact massive two-bond trace

Let $A_N = U C_{L,m} U^\dagger$ and $\Delta_N = A_N - B_N$. The mass-independent unitary of Theorem 1 also scalarizes the massive compression, though A_N is generally not the sine kernel and need not commute with J_N.

Theorem 3 (two-bond Fourier and elliptic identities). Define

$$u_m = \frac{1}{2\pi} \int_0^\pi \frac{m + 1 - \cos k}{E_m(k)} \, dk, \quad v_m = \frac{1}{2\pi} \int_0^\pi \frac{1 - (m + 1) \cos k}{E_m(k)} \, dk.$$

Then u_m and v_m are the intra-cell and inter-cell nearest-neighbor entries of A_N in scalar index order, respectively, while $u_0 = v_0 = 1/\pi$. For every $L \geq 1$,

$$\boxed{\operatorname{Tr}[\Delta_N(-\pi J_N)] = -\pi\left[\frac{4L^2-1}{6}\left(u_m + v_m - \frac{2}{\pi}\right) + \frac{1}{2}(u_m - v_m)\right].}$$

For $m > 0$ put $r = m/(m+2)$, $z = 1 - r^2$, and define complete elliptic integrals with **parameter** z:

$$\mathbf{K}(z) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - z\sin^2\theta}}, \quad \mathbf{E}(z) = \int_0^{\pi/2} \sqrt{1 - z\sin^2\theta}\, d\theta.$$

Then the following identities are exact:

$$\boxed{u_m + v_m = \frac{2}{\pi}\frac{\mathbf{E}(1-r^2) - r^2\mathbf{K}(1-r^2)}{1-r^2}, \qquad u_m - v_m = \frac{2r}{\pi}\frac{\mathbf{K}(1-r^2) - \mathbf{E}(1-r^2)}{1-r^2}.}$$

Proof. For general m, write the Fourier coefficient of P_m as $\frac{1}{2}\delta_{d0} I + x_d(m)\sigma_x + z_d(m)\sigma_z$. Its coefficients satisfy

$$z_d(m) = -\frac{1}{2\pi}\int_0^{\pi} \cos(dk)\frac{m + 1 - \cos k}{E_m(k)}\, dk, \quad x_d(m) = -\frac{i}{2\pi}\int_0^{\pi} \sin(dk)\frac{\sin k}{E_m(k)}\, dk.$$

The same constant orbital rotation and alternating cell gauge show that the intra-cell bond is $-z_0(m) = u_m$. The inter-cell bond is $z_1(m) + ix_1(m)$, which simplifies to v_m because $\sin^2 k - \cos k(m + 1 - \cos k) = 1 - (m+1)\cos k$. At $m = 0$, the formulas in Theorem 1 yield $u_0 = v_0 = 1/\pi$.

Because J_N has only first off-diagonal entries, the trace receives only these two bond contributions. Elementary summation gives

$$\sum_{\substack{1 \leq j < 2L \\ j \text{ odd}}} \frac{j(2L-j)}{2L} = \frac{2L^2+1}{6}, \qquad \sum_{\substack{1 \leq j < 2L \\ j \text{ even}}} \frac{j(2L-j)}{2L} = \frac{L^2-1}{3}.$$

Thus the trace equals $-2\pi\big[\frac{2L^2+1}{6}(u_m - 1/\pi) + \frac{L^2-1}{3}(v_m - 1/\pi)\big]$, algebraically equivalent to the boxed two-bond formula.

Finally, $E_m(k) = (m+2)\sqrt{1 - z\cos^2(k/2)}$. Taking sums and differences of the original Fourier integrals gives

$$u_m + v_m = \frac{m+2}{2\pi}\int_0^{\pi} \frac{1 - \cos k}{E_m(k)}\, dk, \qquad u_m - v_m = \frac{m}{2\pi}\int_0^{\pi} \frac{1 + \cos k}{E_m(k)}\, dk.$$

Set $\theta = (\pi - k)/2$. The integrals reduce to the standard identities $\int \sin^2\theta/\sqrt{1 - z\sin^2\theta} = (\mathbf{K} - \mathbf{E})/z$ and $\int \cos^2\theta/\sqrt{1 - z\sin^2\theta} = (\mathbf{E} - (1-z)\mathbf{K})/z$, yielding exactly the asserted forms. The $m = 0$ case follows by continuity. □

Corollary 4 (unconditional fixed-μ two-bond finite part). Fix $\mu > 0$, set $m = \mu/L$, and let $\mathcal{P}_L = \operatorname{Tr}[\Delta_N(-\pi J_N)]$. Then

$$\boxed{\mathcal{P}_L = \frac{\mu^2}{6}\log L + \mu^2\left(\frac{1}{6}\log\frac{8}{\mu} - \frac{1}{4}\right) + \mathcal{O}_\mu\left(\frac{\log L}{L}\right).}$$

Proof. For $r \downarrow 0$, put $\ell = \log(4/r)$. The classical complementary-modulus expansions (DLMF §19.12) yield

$$\mathbf{K}(1-r^2) = \ell + \mathcal{O}(r^2\ell), \qquad \mathbf{E}(1-r^2) = 1 + \frac{r^2}{2}(\ell - \tfrac{1}{2}) + \mathcal{O}(r^4\ell).$$

It follows that

$$u_m + v_m = \frac{2}{\pi}\left[1 + r^2\left(\frac{3}{4} - \frac{\ell}{2}\right) + \mathcal{O}(r^4\ell)\right], \quad u_m - v_m = \frac{2r}{\pi}\left[\ell - 1 + \mathcal{O}(r^2\ell)\right].$$

For $r = \mu/(2L+\mu)$, one has $(4L^2-1)r^2 = \mu^2 + \mathcal{O}_\mu(L^{-1})$, $r\ell = \mathcal{O}_\mu((\log L)/L)$, and $\ell = \log L + \log(8/\mu) + \mathcal{O}_\mu(L^{-1})$. Inserting these in Theorem 3 gives the result. No estimates for the nonlinear, modular, or entropy branches are used. □

4. Consequences and explicit limits of the derivation

The scalar reference geometry, its Jacobi commutation, and the exact Fourier-to-elliptic reduction of $\mathcal{P}_L$ follow unconditionally from the specified infinite-chain Wilson symbol. The finite relative-entropy identity

$$\mathcal{D}_L = \mathcal{P}_L + \mathcal{Q}_N + \mathcal{F}_L - \Delta S_L$$

requires separate analyses of $\mathcal{Q}_N$, $\mathcal{F}_L$, and ΔS_L to yield a full asymptotic theorem. Their boundedness, assembled in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*, establishes the leading coefficient at fixed positive mass; convergence to a finite part requires stronger control. In particular, the Coulomb/reference-mode occupancy identities, all-degree comparison hypotheses where applicable, and compatible mass-dependent limiting quantifiers are not established by these results. The logarithmic coefficient of $\mathcal{P}_L$ alone must not be identified with a proved coefficient of $\mathcal{D}_L$. This paper concerns finite compressions of full-line projectors and makes no equivalence claim for projectors of finite open-chain Hamiltonians.

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Finite-Interval Compression and Transverse Control

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Abstract

We examine compression of the full-line projector relations for the occupied-band covariances of a one-dimensional Wilson–Dirac chain. The full-line squared-difference and transverse identities imply positive finite-interval inequalities with exact boundary-leakage remainders. A stronger transverse domination follows, together with an explicit noncommutation example showing why the corresponding full-line operator products cannot be transported unchanged to finite intervals.

Operator setting

Let P_m and P_0 be the full-line occupied-band projections for the Wilson–Dirac Bloch symbol

$$h_m(k) = \sin k\,\sigma_x + (1 + m - \cos k)\sigma_z, \quad P_m(k) = \frac{1}{2}\left(I - \frac{h_m(k)}{\sqrt{m^2 + 4(1+m)\sin^2(k/2)}}\right).$$

Fix an interval of L two-orbital cells, with projection χ, $N = 2L$, $\mu > 0$, and $m = \mu/L$. After a common interval-preserving rotation and staggering, the full-line covariance difference is $T = P_m - P_0 = S + D$, where

$$S = -\frac{g_r}{2}Q_0, \qquad D = \frac{a_r}{2}Q_\perp, \qquad G = \frac{g_r}{2}I,$$

$$r = \frac{\mu}{N+\mu}, \qquad u_r = \frac{|\sin(k/2)|}{\sqrt{\sin^2(k/2) + r^2\cos^2(k/2)}}, \quad g_r = 1 - u_r, \quad a_r = \frac{r\cos(k/2)}{\sqrt{\sin^2(k/2) + r^2\cos^2(k/2)}}.$$

The Hermitian involutions $Q_0, Q_\perp$ anticommute. In particular, the **full-line** identities are

$$T^2 = G, \qquad D^2 = G - G^2, \qquad 0 \le G \le \frac{1}{2}I.$$

Set $\Delta_N = \chi T\chi$, $D_N = \chi D\chi$, and $G_N = \chi G\chi$, as operators on $\operatorname{ran}\chi$.

Theorem: exact boundary remainders

For every positive integer L and $\mu > 0$,

$$\boxed{G_N - \Delta_N^2 = \chi T(I-\chi)T\chi \ge 0,}$$

$$\boxed{G_N - G_N^2 - D_N^2 = \chi G(I-\chi)G\chi + \chi D(I-\chi)D\chi \ge 0.}$$

Therefore,

$$\boxed{0 \le \Delta_N^2 \le G_N \le \tfrac{1}{2} I, \quad 0 \le D_N^2 \le G_N - G_N^2 \le G_N.}$$

Taking the finite trace in the first relation gives the exact leakage formula

$$\operatorname{Tr} G_N - \|\Delta_N\|_{\mathrm{HS}}^2 = \|(I-\chi)T\chi\|_{\mathrm{HS}}^2.$$

Proof. For any bounded self-adjoint X, insertion of $I = \chi + (I-\chi)$ gives

$$\chi X^2 \chi - (\chi X \chi)^2 = \chi X (I-\chi) X \chi \ge 0.$$

Apply this identity to T and use $T^2 = G$ for the first formula. For the second, apply it to both G and D, obtaining

$$G_N^2 = \chi G^2 \chi - \chi G (I-\chi) G \chi, \quad D_N^2 = \chi D^2 \chi - \chi D (I-\chi) D \chi.$$

Substitute the full-line equality $G - G^2 - D^2 = 0$. Every remainder is a positive square. Because $0 \le G_N \le I/2$, the remaining order comparisons follow. The trace identity is the Hilbert–Schmidt norm identity for $(I-\chi)T\chi$. □

Proposition: the compressed majorant need not commute with the compressed reference

Let $B_N = \chi P_0 \chi$ in the scalarized basis. At $N = 4$, define

$$c_1 = \frac{1}{\pi} \int_0^\pi g_r(k) \cos k \, dk.$$

The scalarized sine kernel B_4 and the same-parity Toeplitz matrix G_4 satisfy

$$\boxed{[G_4, B_4]_{0,1} = -\frac{2c_1}{3\pi} \neq 0.}$$

Indeed $g_r(k) = 1 - (1 + r^2 \cot^2(k/2))^{-1/2}$ decreases strictly over $(0, \pi)$, so integration by parts shows $c_1 = -\pi^{-1} \int_0^\pi g_r'(k) \sin k \, dk > 0$. The matrix formula follows by direct multiplication of the sine-kernel entries $B_{ij} = \sin[\pi(i-j)/2]/[\pi(i-j)]$ off the diagonal and the parity-supported Fourier coefficients of G_4. □

The full-line $[G, P_0] = 0$ is nevertheless sufficient for the Wilson staggered-pairing inequality: that proof decomposes the interval-supported vector using the full-line projector before evaluating the compressed expectation. The finite model does not require the false identity $[G_N, B_N] = 0$.

Scientific scope and acknowledgments

The theorem pertains to the specified infinite-chain band projections and finite interval compressions, for every $L \ge 1$ and $\mu > 0$. No reference-modular, Coulomb, all-degree nonlinear, or cross-boundary entropy limit follows solely from these identities. The leading relative-entropy law follows only after combining these identities with the separate bounded nonlinear, boundary-entropy and weighted modular estimates, as done in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law.*

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Fourier-Space Boundary Defects for Wilson-Dirac Covariances

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Abstract

For the compression of a self-adjoint, translation-invariant operator to a finite interval, we prove a Fourier-coefficient identity for the exact Hilbert–Schmidt boundary defect. It gives a positive spatial-separation formula and discrete monotonicity and concavity with respect to interval length when the operator is fixed. Applied to the Wilson–Dirac occupied-band projector difference, this formulation gives another derivation of the boundary-leakage identity. For the continuum scaling function, we obtain global derivative bounds and a higher-order large-mass expansion of the signed exponential remainder. The interval-length scaling limit remains dependent on the separately established Hilbert–Schmidt convergence result.

1. A general Fourier-coefficient identity

Let T be a bounded self-adjoint translation-invariant operator on $\ell^2(\mathbb{Z};\mathbb{C}^q)$, with block kernel $T_d \in \mathbb{C}^{q\times q}$:

$$(Tf)_a = \sum_{b\in\mathbb{Z}} T_{a-b} f_b, \qquad T_{-d} = T_d^*.$$

Here the action is initially interpreted using its bounded Fourier multiplier; the sum of squared Frobenius norms $\sum_d \|T_d\|_F^2$ is finite by Plancherel. Let χ_L be projection onto sites $0,\dots,L-1$, and let $Q_L = I - \chi_L$.

Theorem 1 (exact Fourier boundary defect). For every positive integer L,

$$\boxed{\operatorname{Tr}(\chi_L T^2 \chi_L) - \|\chi_L T \chi_L\|_{\mathrm{HS}}^2 = \|Q_L T \chi_L\|_{\mathrm{HS}}^2 = \sum_{d\in\mathbb{Z}} \min\{|d|, L\}\, \|T_d\|_F^2.} \tag{1}$$

Proof. Translation invariance and self-adjointness give

$$\operatorname{Tr}(\chi_L T^2 \chi_L) = L \sum_{d\in\mathbb{Z}} \|T_d\|_F^2.$$

The number of ordered pairs (a,b) with $a,b \in \{0,\dots,L-1\}$ and $a-b=d$ is $L-|d|$ when $|d|<L$, and zero otherwise. Consequently,

$$\|\chi_L T \chi_L\|_{\mathrm{HS}}^2 = \sum_{|d|<L} (L-|d|) \|T_d\|_F^2.$$

Subtracting gives the last expression in (1). Independently, inserting $I = \chi_L + Q_L$ between the two copies of T gives

$$\chi_L T^2 \chi_L - (\chi_L T \chi_L)^2 = \chi_L T Q_L T \chi_L = (Q_L T \chi_L)^*(Q_L T \chi_L).$$

Its trace proves the middle equality and positivity. □

Corollary 2 (discrete concavity for a fixed operator). Write $\mathfrak{L}_L(T)$ for the quantity in (1), and put $\mathfrak{L}_0(T) = 0$. For fixed T,

$$\boxed{\mathfrak{L}_{L+1}(T) - \mathfrak{L}_L(T) = \sum_{|d|\geq L+1} \|T_d\|_F^2 \geq 0,} \tag{2}$$

$$\boxed{\mathfrak{L}_{L+2}(T) - 2\mathfrak{L}_{L+1}(T) + \mathfrak{L}_L(T) = -\sum_{|d|=L+1} \|T_d\|_F^2 \leq 0.} \tag{3}$$

Both follow by taking first and second differences of $\min(L, |d|)$. The operator must be fixed when changing L; neither assertion automatically applies to a sequence of operators whose masses vary with interval length.

2. Wilson–Dirac specialization

For the negative-energy projector of the original two-orbital Wilson–Dirac Bloch symbol,

$$h_m(k) = \sin k\, \sigma_x + (1 + m - \cos k)\sigma_z, \quad P_m(k) = \tfrac{1}{2}\left[I - \frac{h_m(k)}{\sqrt{m^2 + 4(1+m)\sin^2(k/2)}}\right],$$

put $T^{(m)} = P_m - P_0$ as a full-line operator, and denote its Fourier coefficients by $T_d^{(m)}$. The interval compression is precisely the covariance difference used in the research program. Hence Theorem 1 gives the exact all-length identity

$$\boxed{\operatorname{Tr} G_L - \|\Delta_L\|_{\mathrm{HS}}^2 = \sum_{d\in\mathbb{Z}} \min(L, |d|)\, \|T_d^{(m)}\|_F^2,} \tag{4}$$

where $G_L = \chi_L (T^{(m)})^2 \chi_L$. This holds for every $m > 0$, including the scaling choice $m = \mu/L$. The representation makes clear that only nonzero spatial separations contribute to leakage. It is independent of the finite Jacobi basis and the modular spectral approximation.

For fixed $\mu > 0$, the separately established Hilbert–Schmidt convergence statement and the exact trace limit yield

$$\lim_{L\to\infty} \mathfrak{L}_L(T^{(\mu/L)}) = \Lambda(\mu) = \frac{1}{\pi^2}\int_0^\infty \min(t, \mu)\Phi(t)\, dt,$$

with $\Phi(t) = K_0(t)^2 + [K_1(t) - t^{-1}]^2$. Equation (4) supplies an alternative exact finite representation, not a replacement proof of the already stated finite-to-continuum sampling theorem.

Continuum Fourier normalization. With the inverse Fourier convention $\check{F}(x) = (2\pi)^{-1}\int_{\mathbb{R}} e^{iux} F(u)\, du$, define

$$F_\mu(u) = -\frac{1}{2}\left[\left(\frac{u}{\sqrt{u^2+\mu^2}} - \operatorname{sgn} u\right)\sigma_x + \frac{\mu}{\sqrt{u^2+\mu^2}}\sigma_z\right].$$

The Pauli trace identity gives $\|F_\mu(u)\|_F^2 = 1 - |u|/\sqrt{u^2+\mu^2}$ and $(2\pi)^{-1}\int_{\mathbb{R}} \|F_\mu(u)\|_F^2 du = \mu/\pi$. The standard modified-Bessel Fourier transforms give, for $x \neq 0$,

$$\|\check{F}_\mu(x)\|_F^2 = \frac{1}{2\pi^2}\left[\mu^2 K_0(\mu|x|)^2 + \left(\mu K_1(\mu|x|) - \frac{1}{|x|}\right)^2\right].$$

Plancherel, evenness and the substitution $t = \mu|x|$ consequently prove

$$\boxed{\int_0^\infty \Phi(t)\,dt = \pi.}$$

This normalization makes the identity $\Lambda(\mu) = \mu/\pi - H(\mu) = \pi^{-2}\int_0^\infty \min(t,\mu)\Phi(t)dt$ explicit; the finite-to-continuum limit itself is supplied by the separate Hilbert–Schmidt scaling theorem.

3. Global differential bounds and a sharper remainder expansion

Let

$$C(\mu) = \frac{\log(\mu/2)+\gamma+1}{\pi^2}, \qquad W(t) = \frac{2K_1(t)}{t} - K_0(t)^2 - K_1(t)^2.$$

Standard Bessel identities imply $0 < K_0(t) < K_1(t) < 1/t$, so $W(t) > 0$ and

$$\Phi(t) = \frac{1}{t^2} - W(t), \qquad 0 < \Phi(t) < \frac{1}{t^2}. \tag{5}$$

The normalization constant is fixed directly: the Bessel derivative identities give

$$\frac{d}{dt}\{-tK_0(t)K_1(t) + 2K_0(t) + \log t\} = t\Phi(t).$$

The expression in braces tends to $\log 2 - \gamma$ as $t \downarrow 0$, and differs from $\log t$ by an exponentially decaying term at infinity. Combining this antiderivative with $\int_0^\infty \Phi(t)dt = \pi$ proves the exact signed-remainder representation

$$R(\mu) := \Lambda(\mu) - C(\mu) = \frac{1}{\pi^2}\int_\mu^\infty (t-\mu)W(t)\,dt > 0. \tag{6}$$

Proposition 3 (derivative bounds). For all $\mu > 0$,

$$\boxed{0 < \Lambda'(\mu) < \frac{1}{\pi^2\mu}, \qquad -\frac{1}{\pi^2\mu^2} < \Lambda''(\mu) < 0.} \tag{7}$$

Moreover $R'(\mu) < 0$, $R''(\mu) > 0$. Indeed,

$$\Lambda'(\mu) = \frac{1}{\pi^2}\int_\mu^\infty \Phi(t)dt, \quad \Lambda''(\mu) = -\frac{\Phi(\mu)}{\pi^2}, \quad R'(\mu) = -\frac{1}{\pi^2}\int_\mu^\infty W(t)dt, \quad R''(\mu) = \frac{W(\mu)}{\pi^2}.$$

The strictly positive inequalities follow from (5) and (6). □

Proposition 4 (three-term exponential equivalent). As $\mu \to \infty$,

$$\boxed{R(\mu) = \frac{\sqrt{2\pi}}{\pi^2}\frac{e^{-\mu}}{\mu^{3/2}}\left[1 - \frac{21}{8\mu} + \frac{1185}{128\mu^2} + O(\mu^{-3})\right].} \tag{8}$$

Proof. The standard Bessel expansion gives

$$K_1(t) = \sqrt{\frac{\pi}{2t}}e^{-t}\left[1 + \frac{3}{8t} - \frac{15}{128t^2} + O(t^{-3})\right].$$

The squares $K_0(t)^2+K_1(t)^2 = O(e^{-2t}/t)$ make an exponentially smaller contribution. Therefore

$$W(t) = \sqrt{2\pi}\, e^{-t} t^{-3/2} \left[1 + \frac{3}{8t} - \frac{15}{128t^2} + O(t^{-3})\right] + O(e^{-2t}/t).$$

In (6), substitute $t = \mu + s$, expand uniformly under the integrable majorant se^{-s} and use $\int_0^\infty s^j e^{-s} ds = j!$. The μ^{-1} coefficient is $3/8 - (3/2)\cdot 2 = -21/8$. The μ^{-2} coefficient is

$$\frac{15}{8}\cdot 6 - \frac{15}{16}\cdot 2 - \frac{15}{128} = \frac{1185}{128}.$$

The uniform higher-order remainder and exponentially small squared-Bessel contribution give the stated error. □

4. Scope

These results apply to a specified translation-invariant covariance difference, together with the continuum scaling function in *Hilbert-Schmidt Scaling Limit for Wilson-Dirac Covariances*. They are not statements about the full quasifree relative entropy. That comparison additionally uses nonlinear, modular and cross-boundary estimates: the complete fixed-μ assembly is Theorem 1 and Section 7 of *Finite Part of Wilson–Dirac Relative Entropy*, with sharpened modular and nonlinear bounds in *Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals*. In particular, the discrete concavity in (3) is **only** a fixed-operator statement.

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Fourth-Moment Control of the Wilson Nonlinear Correction

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Abstract

For the mass-scaled Wilson–Dirac interval covariance, we derive a vanishing bound for the transverse nonlinear correction, sharpening a square-root-logarithmic upper bound. The argument combines the proven positive spectral-majorant tail estimate with the sine-kernel/Jacobi second moment. Specifically, the transverse contribution is $O_\mu(\sqrt{\log N/N})$, while the longitudinal contribution remains $O_\mu(1)$. Consequently the complete nonlinear correction is uniformly bounded at fixed scaled mass. Together with a separately established cross-boundary entropy bound, this reduces the leading relative-entropy problem to its exact reference-modular residual, without proving the latter vanishes or has a prescribed asymptotic order.

1. Model and inputs

Let $N = 2L$ and $m = \mu/L$ with fixed $\mu > 0$. Consider the interval compression of the occupied-band projection of

$$h_m(k) = \sin k\, \sigma_x + (1 + m - \cos k)\sigma_z.$$

Under the established massless scalarization, the reference covariance is B_N, with commuting Jacobi matrix J_N. Write $W_N = B_N(I - B_N)$, and decompose the perturbation as $\Delta_N = S_N + D_N$. Let v_j be common eigenvectors of B_N and J_N, with eigenvalues $b_j \in (0, 1)$ and λ_j; put $q_j = \langle v_j, G_N v_j\rangle \geq 0$.

The already proved Wilson spectral-majorant theorem supplies

$$|\langle v_j, S_N v_j\rangle| \leq q_j, \qquad |\langle v_j, D_N v_j\rangle| \leq 2\sqrt{q_j b_j(1 - b_j)},$$
$$\sum_{|\lambda_j| > t} q_j \leq \frac{a_\mu}{t} + \frac{b_\mu}{t^2}, \qquad a_\mu = \sqrt{\frac{2}{15}}\frac{\mu^2}{\pi}, \quad b_\mu = \frac{\mu}{3\pi}, \qquad t > 0.$$

The Jacobi normalization ensures $|\lambda_j| \leq N/2$. The exact half-filled sine-kernel boundary identity implies

$$0 \leq M_{2,N} := \mathrm{Tr}(J_N^2 W_N) \leq \frac{32}{\pi^2}(H_N + 1),$$

and the refined exterior-lattice computation proves $M_{2,N} = (3\pi^2)^{-1} \log N + O(1)$.

Define $\eta(t) = \pi t\, {}_3F_2(\frac{1}{4}, \frac{1}{2}, \frac{3}{4}; 1, \frac{3}{2}; 4t^2)$ and $V_N = -N[\eta(J_N/N) - \pi J_N/N]$. The positive-coefficient hypergeometric series gives

$$|\eta(t) - \pi t| \leq C_\eta |t|^3 \quad (|t| \leq \tfrac{1}{2}), \qquad C_\eta = 4\pi\big[{}_3F_2(\tfrac{1}{4}, \tfrac{1}{2}, \tfrac{3}{4}; 1, \tfrac{3}{2}; 1) - 1\big].$$

All inputs are specific to the Wilson–Dirac/sine-kernel model and do not assume a reference-modular comparison.

2. A fourth-moment bound for the positive majorant

Lemma 1. For every even $N \geq 2$,

$$\boxed{\mathrm{Tr}(G_N J_N^4) = \sum_j q_j |\lambda_j|^4 \leq \frac{a_\mu N^3}{6} + \frac{b_\mu N^2}{2}.}$$

Proof. The diagonal representation holds because J_N^4 is diagonal in the common reference eigenbasis, regardless of whether G_N commutes with J_N. For the positive measure $\nu_N = \sum_j q_j \delta_{|\lambda_j|}$, layer-cake integration and the Jacobi spectral radius give

$$\begin{aligned}\mathrm{Tr}(G_N J_N^4) &= 4 \int_0^{N/2} t^3 \nu_N((t,\infty))\, dt \\ &\leq 4 \int_0^{N/2} (a_\mu t^2 + b_\mu t)\, dt = \frac{a_\mu N^3}{6} + \frac{b_\mu N^2}{2}.\end{aligned}$$

The singular appearance of the spectral-tail bound near $t = 0$ causes no problem: the integrated majorant is finite. □

3. Improved transverse estimate

Theorem 2. Let $\mathcal{Q}_N^{\mathrm{trans}} = \mathrm{Tr}(D_N V_N)$. Then

$$\boxed{|\mathcal{Q}_N^{\mathrm{trans}}| \leq 2C_\eta \sqrt{\left(\frac{a_\mu}{6N} + \frac{b_\mu}{2N^2}\right) M_{2,N}} = O_\mu\left(\sqrt{\frac{\log N}{N}}\right).}$$

In particular $\mathcal{Q}_N^{\mathrm{trans}} \to 0$ as $N \to \infty$ at each fixed $\mu > 0$.

Proof. In the reference eigenbasis, V_N has diagonal entries w_j with $|w_j| \leq C_\eta |\lambda_j|^3 / N^2$. Using the pairing inequality followed by Cauchy–Schwarz with the **fourth positive-majorant moment**, not merely its total mass, yields

$$\begin{aligned}|\mathcal{Q}_N^{\mathrm{trans}}| &\leq \frac{2C_\eta}{N^2} \sum_j |\lambda_j|^3 \sqrt{q_j b_j (1 - b_j)} \\ &\leq \frac{2C_\eta}{N^2} \left(\sum_j q_j \lambda_j^4\right)^{1/2} \left(\sum_j \lambda_j^2 b_j (1 - b_j)\right)^{1/2} \\ &= \frac{2C_\eta}{N^2} \sqrt{\mathrm{Tr}(G_N J_N^4)\, \mathrm{Tr}(J_N^2 W_N)}.\end{aligned}$$

Apply Lemma 1 and the established Jacobi moment bound. □

The finite-length estimate is explicit; the sharp asymptotic coefficient for $M_{2,N}$ is not required to obtain the rate, only its $O(\log N)$ upper bound.

4. Bounded nonlinear correction and modular reduction

The established longitudinal estimate is

$$|\mathcal{Q}_N^{\mathrm{long}}| \leq C_\eta \left(\frac{3a_\mu}{8} + \frac{3b_\mu}{2N}\right).$$

Therefore

$$\boxed{|\mathcal{Q}_N| \le C_\eta \left(\frac{3a_\mu}{8} + \frac{3b_\mu}{2N}\right) + 2C_\eta \sqrt{\left(\frac{a_\mu}{6N} + \frac{b_\mu}{2N^2}\right) M_{2,N}} = O_\mu(1).}$$

The exact two-bond asymptotic and finite quasifree relative-entropy decomposition are

$$\mathcal{P}_L = \frac{\mu^2}{6} \log L + O_\mu(1), \qquad \mathcal{D}_L = \mathcal{P}_L + \mathcal{Q}_N + \mathcal{F}_L - \Delta S_L,$$

where $\mathcal{F}_L = \mathrm{Tr}[(A_N - B_N)E_N]$ is the exact reference-modular residual. Theorem 1 of *Uniform Square-Root-Mass Control of Wilson–Dirac Boundary Covariances*, with its entropy argument in section 5, proves $\Delta S_L = O_\mu(1)$ for this Wilson scaling. Combining that imported bound with the nonlinear result of this paper yields

$$\boxed{\mathcal{D}_L = \frac{\mu^2}{6} \log L + \mathcal{F}_L + O_\mu(1).}$$

This reduction alone does not prove $\mathcal{F}_L = o(\log L)$, convergence to a finite constant, or a proposed inverse-logarithmic convergence rate. The separate theorem in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* proves $\mathcal{F}_L = O_\mu(1)$, sufficient for the leading law. Theorem 3 of *Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals* proves $\|E_N\| \le 7$ for every even $N \ge 2$, under the exact concentration connection specified there. This later result is separate from the reduction proved above.

5. Evidence and scope

The fourth-moment estimate and theorem are exact mathematical consequences of the stated majorant and sine-kernel results. Independent finite arithmetic compares the improved bound with the earlier bound for eight Wilson configurations; all satisfy the stronger estimate. These tests are consistency checks, not the proof. The claimed rate holds at fixed μ only, with no uniform assertion for μ varying with N.

External mathematical context

- V. Eisler and I. Peschel, "Analytical results for the entanglement Hamiltonian of a free-fermion chain," *J. Phys. A* **50** (2017) 284003, doi:10.1088/1751-8121/aa76b5.
- D. Slepian, "Prolate spheroidal wave functions, Fourier analysis, and uncertainty. V: The discrete case," *Bell System Technical Journal* **57** (1978), 1371–1430, doi:10.1002/j.1538-7305.1978.tb02104.x.
- S. Karnik, J. Romberg and M. A. Davenport, "Improved bounds for the eigenvalues of prolate spheroidal wave functions and discrete prolate spheroidal sequences," *Applied and Computational Harmonic Analysis* **55** (2021), 97–128, doi:10.1016/j.acha.2021.04.002.

These references support the classical concentration-operator setting; they do not independently prove the new Wilson-specific fourth-moment consequence.

Acknowledgments and contributions

Kirtus Justus directed the research. ChatGPT (OpenAI) developed and audited this fourth-moment derivation and its computational corroboration. Claude (Anthropic), Gemini (Google), Grok (SpaceXAI), and DeepSeek (High-Flyer) are acknowledged as broader research contributors without attribution of this derivation to them. There has been no external human peer review or institutional endorsement.

A Hilbert–Schmidt Scaling Limit for Finite-Interval Wilson–Dirac Covariances

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Abstract. For the restrictions of two infinite-chain Wilson–Dirac occupied-band projections to an interval of L cells, we establish a Hilbert–Schmidt scaling limit when the massive parameter is $m = \mu/L$ with fixed $\mu > 0$. The limiting integral operator on $L^2(0,1;\mathbb{C}^2)$ has an explicitly computable modified-Bessel kernel. A dimension-free lower bound for the finite quasifree relative entropy then gives a strictly positive limiting lower benchmark, without a spectral-mode localization hypothesis. The benchmark is an increasing, strictly convex function of μ; its small-mass expansion and large-mass slope are explicit. The arguments use only the occupied-band symbol, Fourier–Plancherel theory, and finite-matrix convexity.

Keywords: Wilson–Dirac chain; quasifree fermions; Hilbert–Schmidt convergence; quantum relative entropy; Bessel functions; Toeplitz compression.

1. Model and result

Write σ_x, σ_z for the Pauli matrices and fix $\mu > 0$. For $m \geq 0$ and $k \in [-\pi, \pi]$, put

$$h_m(k) = \sin k\, \sigma_x + (1 + m - \cos k)\, \sigma_z, \qquad E_m(k) = \sqrt{m^2 + 4(1+m)\sin^2(k/2)}.$$

The infinite-chain occupied-band multiplier is $P_m(k) = \frac{1}{2}(I - h_m(k)/E_m(k))$. Its value at $k = m = 0$ may be specified arbitrarily, since a single Fourier point is immaterial. Let $C_{L,m}$ be the block Toeplitz compression to L consecutive two-orbital cells,

$$(C_{L,m})_{jk} = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{i(j-k)\theta} P_m(\theta)\, d\theta, \qquad 0 \leq j, k < L.$$

We compare the massive and massless compressions at $m = \mu/L$:

$$A_L = C_{L,\mu/L}, \qquad B_L = C_{L,0}, \qquad \Delta_L = A_L - B_L.$$

The finite compressed covariances satisfy $0 < C_{L,m} < I$ for every $m \geq 0$. To see this, suppose that a nonzero finitely supported vector has zero quadratic form under $C_{L,m}$ or $I - C_{L,m}$. Its nonzero vector-valued Laurent polynomial $v(z)$, with $z = e^{ik}$, would then satisfy $h_m(k)v(z) = \pm E_m(k)v(z)$ almost everywhere. A nonzero component forces $E_m(k)$ to agree there with a rational function of z. But

$$E_m(k)^2 = m^2 + 2(1+m) - (1+m)(z + z^{-1})$$

has a pole of odd order at $z = 0$, whereas the square of a rational function has only poles of even order. This contradiction proves both strict inequalities.

The trace convention $\|M\|_2^2 = \mathrm{Tr}(M^*M)$ is **un-normalized**, and the logarithms in the relative entropy are natural. Define

$$\mathcal{D}_L(\mu) = \mathrm{Tr}\{A_L(\log A_L - \log B_L) + (I - A_L)[\log(I - A_L) - \log(I - B_L)]\}.$$

This is the relative entropy of the corresponding finite gauge-invariant quasifree states.

Theorem 1 (Covariance scaling and relative-entropy bound). For fixed $\mu > 0$, the matrices Δ_L have a Hilbert–Schmidt continuum limit after the canonical piecewise-constant embedding of the L cells into $(0,1)$. Its

kernel for $s = x - y \neq 0$ is

$$\mathcal{K}_\mu(s) = \frac{i}{2\pi}\left[\frac{1}{s} - \mu\,\mathrm{sgn}(s)K_1(\mu|s|)\right]\sigma_x - \frac{\mu}{2\pi}K_0(\mu|s|)\sigma_z. \tag{1}$$

In particular,

$$\boxed{\lim_{L\to\infty}\|\Delta_L\|_2^2 = H(\mu) = \frac{1}{\pi^2}\int_0^1 (1-s)\left\{\mu^2 K_0(\mu s)^2 + \left[\frac{1}{s} - \mu K_1(\mu s)\right]^2\right\} ds.} \tag{2}$$

The associated relative entropy obeys

$$\boxed{\liminf_{L\to\infty}\mathcal{D}_L(\mu) \geq 2H(\mu) > 0.} \tag{3}$$

The additional estimates in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* establish the relative entropy's leading logarithmic coefficient with bounded remainder. Equation (3) is a separate all-mass, fixed-μ lower benchmark that can be evaluated without using that theorem.

2. Proof of the Hilbert–Schmidt limit

Lemma 2 (Rescaled Fourier multipliers). Set

$$F_L(t) = \mathbf{1}_{[-\pi L,\pi L]}(t)[P_{\mu/L}(t/L) - P_0(t/L)].$$

As $L\to\infty$, $F_L \to F_\mu$ in $L^2(\mathbb{R};M_2(\mathbb{C}))$, where, for $t\neq 0$,

$$F_\mu(t) = -\frac{1}{2}\left\{\left[\frac{t}{\sqrt{t^2+\mu^2}} - \mathrm{sgn}\,t\right]\sigma_x + \frac{\mu}{\sqrt{t^2+\mu^2}}\sigma_z\right\}. \tag{4}$$

Moreover $\|tF_L\|^2_{L^2(\mathbb{R};\mathrm{F})} = O_\mu(L)$.

Proof. For fixed $t\neq 0$, use $L\sin(t/L)\to t$ and $L(1-\cos(t/L))\to 0$ in the normalized Bloch vector. The massless σ_x component tends to $\mathrm{sgn}\,t$, while its σ_z component tends to zero, giving (4).

For $k\neq 0$ in the Brillouin zone, the vector $h_m(k)$ has length $E_m(k)\geq E_0(k) = 2|\sin(k/2)| \geq 2|k|/\pi$. Differentiating its normalized direction with respect to m gives a Euclidean norm at most $1/E_m(k)$. The mean-value bound for the difference of the two rank-one spectral projections, supplemented by the elementary uniform bound for differences of projections, therefore gives

$$\|F_L(t)\|_{\mathrm{F}} \leq C\min\left\{1, \frac{\mu}{|t|}\right\} \quad (t\neq 0), \tag{5}$$

with C independent of L and μ. The right side is square-integrable in t for fixed $\mu>0$ and also bounds F_μ. Dominated convergence proves the L^2 assertion. Finally $|t|\|F_L(t)\|_{\mathrm{F}} \leq C\mu$ on its support, an interval of length $2\pi L$, so the stated weighted L^2 bound follows. □

Let

$$K_L(s) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{its}F_L(t)\,dt, \qquad K_\mu = \mathcal{F}^{-1}F_\mu$$

using the same Fourier normalization. Plancherel and Lemma 2 imply

$$\|K_L - K_\mu\|_{L^2(\mathbb{R};\mathrm{F})} \longrightarrow 0, \qquad \|K_L'\|^2_{L^2(\mathbb{R};\mathrm{F})} = O_\mu(L). \tag{6}$$

In particular $K_L \in H^1(\mathbb{R};M_2)$ has a continuous representative, so its lattice samples are well-defined.

Lemma 3 (Sampling does not change the limit). Map the standard basis of $\mathbb{C}^L$ into $L^2(0,1)$ by $e_j \mapsto \sqrt{L}\,\mathbf{1}_{[j/L,(j+1)/L)}$. Under its tensor product with $I_{\mathbb{C}^2}$, the finite covariance difference becomes an integral operator with kernel

$$M_L(x,y) = K_L\left(\frac{\lfloor Lx\rfloor - \lfloor Ly\rfloor}{L}\right).$$

Then $\|M_L - K_\mu(x-y)\|_{L^2((0,1)^2;\mathrm{F})} \to 0$.

Proof. The exact block Fourier formula, after $t = L\theta$, gives $(\Delta_L)_{jk} = L^{-1}K_L((j-k)/L)$, which establishes the displayed kernel. If x and y belong to the jth and kth cells, let $s_0 = (j-k)/L$ and $s = x-y$. Then $|s-s_0| \le 1/L$, and the fundamental theorem of calculus and Cauchy–Schwarz give

$$\|K_L(s) - K_L(s_0)\|_{\mathrm{F}}^2 \le \frac{1}{L}\int_{s_0-1/L}^{s_0+1/L} \|K_L'(u)\|_{\mathrm{F}}^2 du.$$

Each pair of cells has area L^{-2}; for each difference $d = j-k$, there are $L-|d|$ pairs, and the intervals $[(d-1)/L, (d+1)/L]$ overlap at most threefold. Consequently,

$$\|M_L - K_L(x-y)\|_{L^2((0,1)^2;\mathrm{F})}^2 \le \frac{3}{L^2}\|K_L'\|_{L^2(\mathbb{R};\mathrm{F})}^2 = O_\mu(L^{-1}).$$

Also

$$\int_0^1\int_0^1 \|K_L(x-y) - K_\mu(x-y)\|_{\mathrm{F}}^2 dx\,dy = \int_{-1}^1 (1-|s|)\|K_L(s) - K_\mu(s)\|_{\mathrm{F}}^2 ds \le \|K_L - K_\mu\|_{L^2(\mathbb{R};\mathrm{F})}^2.$$

Equation (6) completes the proof. □

Lemma 4 (Modified-Bessel kernel). For $s \neq 0$, $K_\mu(s)$ is given by (1).

Proof. The cosine transform

$$\int_0^\infty \frac{\cos(ts)}{\sqrt{t^2+\mu^2}} dt = K_0(\mu|s|)$$

and $K_0'(z) = -K_1(z)$ give the Fourier transform of $t/\sqrt{t^2+\mu^2}$ by differentiation. The Fourier transform of $\operatorname{sgn} t$ is $i/(\pi s)$ in the tempered-distribution sense with the present sign convention. Their difference lies in L^2; the singular terms cancel when combined. The σ_z component is the ordinary even cosine transform. Substitution into (4) yields (1). □

Proof of the limit (2). The $K_0(\mu|s|)$ component has a logarithmic singularity at $s = 0$; the coefficient of σ_x vanishes there at rate $O_\mu(s|\log|s||)$. Thus the kernel is locally square-integrable. Orthogonality of the Pauli matrices under the Frobenius inner product yields

$$\|K_\mu(s)\|_{\mathrm{F}}^2 = \frac{1}{2\pi^2}\left\{\left[\frac{1}{s} - \mu\operatorname{sgn}(s)K_1(\mu|s|)\right]^2 + \mu^2 K_0(\mu|s|)^2\right\}.$$

Lemma 3 gives convergence of the embedded operators in Hilbert–Schmidt norm. Integrating over the unit square, using $\int_0^1\int_0^1 f(x-y)dxdy = \int_{-1}^1 (1-|s|)f(s)ds$, proves (2). □

3. Finite-dimensional entropy inequality

Proposition 5 (Sharp covariance Hilbert–Schmidt bound). If $0 < A, B < I$ are finite Hermitian matrices of the same size, then

$$\mathcal{D}(A\|B) \ge 2\operatorname{Tr}(A-B)^2. \tag{7}$$

The coefficient 2 cannot be increased uniformly over all such matrices.

Proof. Let $f(u) = u\log u + (1-u)\log(1-u)$ on $(0,1)$, and put $V = A - B$, $B_t = B + tV$, and $g(t) = \operatorname{Tr} f(B_t)$. Direct rearrangement of the trace entropy gives $\mathcal{D}(A\|B) = g(1) - g(0) - g'(0)$. Diagonalize B_t with eigenvalues $\lambda_i(t)$. Differentiation of a trace spectral function gives

$$g''(t) = \sum_{i,j} |V_{ij}^{(t)}|^2 \begin{cases} \dfrac{f'(\lambda_i) - f'(\lambda_j)}{\lambda_i - \lambda_j}, & \lambda_i \neq \lambda_j, \\ f''(\lambda_i), & \lambda_i = \lambda_j. \end{cases}$$

This divided-difference identity is valid also at degeneracies by continuous extension. Since $f''(u) = 1/[u(1-u)] \geq 4$, every coefficient in the sum is at least 4 by the scalar mean-value theorem. Therefore $g''(t) \geq 4\|V\|_2^2$. Taylor's integral identity gives

$$\mathcal{D}(A\|B) = \int_0^1 (1-t)g''(t)dt \geq 2\|V\|_2^2.$$

For one-dimensional $B = 1/2$ and $A = 1/2 + \epsilon$, the ratio $\mathcal{D}(A\|B)/\epsilon^2 \to 2$ as $\epsilon \to 0$, proving sharpness. □

Proof of (3). Apply (7) to A_L and B_L, and take the lower limit using (2). The strict positivity follows from $K_0(t) > 0$ for $t > 0$, since its squared contribution has positive integral. □

4. Dependence on the mass parameter

Put

$$G(t) = K_0(t)^2 + \left(K_1(t) - \frac{1}{t}\right)^2, \qquad t > 0.$$

The change of variable $t = \mu s$ gives a useful second representation,

$$\boxed{H(\mu) = \frac{1}{\pi^2}\int_0^\mu (\mu - t)G(t)\,dt.} \tag{8}$$

The behavior of K_0 and K_1 at zero shows $G(t) = O(1 + |\log t|^2)$, while $G(t) = t^{-2} + O(e^{-t}t^{-1/2})$ at infinity.

Corollary 6 (Monotonicity and convexity). The function H is twice continuously differentiable on $(0, \infty)$ and

$$H'(\mu) = \frac{1}{\pi^2}\int_0^\mu G(t)dt > 0, \qquad H''(\mu) = \frac{G(\mu)}{\pi^2} > 0. \tag{9}$$

Proposition 7 (Exact integral and large-mass slope). One has

$$\int_0^\infty \left[K_0(t)^2 + \left(K_1(t) - \frac{1}{t}\right)^2\right] dt = \pi, \qquad \boxed{\lim_{\mu\to\infty} \frac{H(\mu)}{\mu} = \frac{1}{\pi}.} \tag{10}$$

Proof. Formula (4), with $\mu = 1$, gives the exact Frobenius square

$$\|F_1(t)\|_{\mathrm{F}}^2 = 1 - \frac{|t|}{\sqrt{t^2+1}}.$$

Its integral over $\mathbb{R}$ is 2, by the elementary antiderivative $t - \sqrt{1+t^2}$ on the positive axis. Fourier–Plancherel thus gives $\int_{\mathbb{R}} \|K_1(s)\|_{\mathrm{F}}^2 ds = 1/\pi$. Inserting the Bessel kernel (1) and using its even Frobenius norm gives the first identity in (10). The expression $H(\mu)/\mu = \pi^{-2}\int_0^\mu (1 - t/\mu)G(t)dt$ converges to $\pi^{-2}\int_0^\infty G(t)dt$ by dominated convergence. □

Proposition 8 (Small-mass expansion). Let $a_\mu = \log(2/\mu) - \gamma$, where γ is Euler's constant. As $\mu \downarrow 0$,

$$\boxed{H(\mu) = \frac{\mu^2}{\pi^2}\left(\frac{a_\mu^2}{2} + \frac{3a_\mu}{2} + \frac{7}{4}\right) + O(\mu^4[1 + |\log\mu|]^2)\,.} \tag{11}$$

Proof. The standard modified-Bessel expansions, uniformly for $0 < s \leq 1$ and small μ, give

$$K_0(\mu s) = a_\mu - \log s + O(\mu^2 s^2[1 + |\log(\mu s)|])\,, \quad \frac{1}{s} - \mu K_1(\mu s) = O(\mu^2 s[1 + |\log(\mu s)|])\,.$$

Use these in (2). The integrable error terms have the stated order. The elementary integrals $\int_0^1 (1-s)ds = 1/2$, $\int_0^1 (1-s)(-\log s)ds = 3/4$, and $\int_0^1 (1-s)(\log s)^2 ds = 7/4$ yield (11). □

The limit $L \to \infty$ throughout this manuscript is taken with μ fixed. Equations (10)–(11) describe the **subsequent** behavior of the limiting function $H(\mu)$ as its argument changes; they require no exchange of the L and μ limits.

5. Numerical diagnostics

The table below compares direct finite-lattice Fourier integration of $\mathrm{Tr}(\Delta_L^2)$ with independent quadrature of the Bessel representation (2). Numerical values are rounded and are not interval enclosures.

μ	$H(\mu)$	$\|\Delta_{16}\|_2^2$	$\|\Delta_{64}\|_2^2$	$\|\Delta_{128}\|_2^2$
0.00625	0.00009107247233	0.00009059369729	0.00009097366525	0.00009102547241
0.05	0.002852564396	0.002817285774	0.002844634318	0.002848706782
0.5	0.084815500230	0.080385074026	0.083727553206	0.084275474084
2	0.471615679441	0.406078605117	0.454191601689	0.462834231123

As a separate check, the finite Frobenius coefficient sum was compared against explicit construction of the complex block Toeplitz matrix, and the matrix-entropy inequality was tested on noncommuting positive contractions. All derivations above are analytical; the numerical diagnostics help expose normalization and Fourier-sign mistakes.

References

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Acknowledgments and contributions

Kirtus Justus is responsible for the research direction, selection of the mathematical problem, and the final scientific claims presented under his byline. ChatGPT (OpenAI) assisted with the derivation of the kernel limit, matrix-entropy inequality, numerical verification, and manuscript drafting. Claude (Anthropic), Gemini (Google), Grok (SpaceXAI), and DeepSeek (High-Flyer) are acknowledged as participants in the broader research workflow; no specific contribution to this derivation is attributed to them without source-level provenance. All mathematical claims remain subject to human review. No institutional endorsement is implied.

Signed Exponential Remainder for Wilson-Dirac Covariance Distances

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Abstract

The continuum Hilbert–Schmidt covariance-distance function associated with the fixed-positive-mass scaling of a one-dimensional Wilson–Dirac chain admits an exact, sign-definite remainder beyond its linear and logarithmic large-parameter terms. The remainder has a strictly positive integral kernel, an explicit global upper bound in terms of the modified Bessel function K_1, and a sharp leading exponential equivalent. The calculation uses Fourier–Plancherel normalization, a Bessel first-moment identity, and classical integral representations. It concerns the covariance-distance scaling function, with the interval-size limit already taken; it entails no asymptotic assertion about the complete quasifree relative entropy.

1. Main result

For $t, \mu > 0$, define

$$G(t) = K_0(t)^2 + \left(K_1(t) - \frac{1}{t}\right)^2, \qquad H(\mu) = \frac{1}{\pi^2}\int_0^\mu (\mu - t)G(t)\,dt,$$

and

$$A(\mu) = \frac{\mu}{\pi} - \frac{\log(\mu/2) + \gamma + 1}{\pi^2}, \qquad W(t) = \frac{2K_1(t)}{t} - K_0(t)^2 - K_1(t)^2.$$

Here γ denotes Euler's constant and K_ν the modified Bessel function of the second kind.

Theorem 1 (exact signed remainder and global quantitative bound). For every $\mu > 0$,

$$\boxed{A(\mu) - H(\mu) = \frac{1}{\pi^2}\int_\mu^\infty (t - \mu)W(t)\,dt, \qquad W(t) > 0 \quad (t > 0).} \tag{1}$$

Consequently,

$$\boxed{0 < A(\mu) - H(\mu) < \frac{2K_1(\mu)}{\pi^2\mu} \qquad (\mu > 0).} \tag{2}$$

The gap $A - H$ is strictly decreasing and strictly convex, and

$$\boxed{A(\mu) - H(\mu) = \frac{\sqrt{2\pi}}{\pi^2}\,\frac{e^{-\mu}}{\mu^{3/2}}\,(1 + O(\mu^{-1})) \quad (\mu \to \infty).} \tag{3}$$

The upper bound (2) is asymptotically sharp: the ratio of its right-hand side to $A - H$ tends to 1.

2. Proof

For completeness, the modified-Bessel recurrences and derivative identities are

$$K_0'(t) = -K_1(t), \qquad K_1'(t) = -K_0(t) - \frac{K_1(t)}{t},$$

and yield the exact first-moment identity

$$\int_0^\mu tG(t)\,dt = \log(\mu/2) + \gamma + 2K_0(\mu) - \mu K_0(\mu)K_1(\mu). \tag{4}$$

Indeed the derivative of the right-hand side is $\mu G(\mu)$; the constant follows from $K_0(t) = -\log(t/2) - \gamma + O(t^2|\log t|)$ and $tK_1(t) = 1 + O(t^2|\log t|)$ as $t \downarrow 0$. For clarity, the normalization is verified directly. The rescaled occupied-band projector difference at parameter one has Fourier multiplier

$$F_1(p) = -\frac{1}{2}\left[\left(\frac{p}{\sqrt{p^2+1}} - \operatorname{sgn} p\right)\sigma_x + \frac{1}{\sqrt{p^2+1}}\sigma_z\right].$$

Using $\mathrm{Tr}(\sigma_a\sigma_b) = 2\delta_{ab}$, its squared Frobenius norm is

$$\|F_1(p)\|_F^2 = 1 - \frac{|p|}{\sqrt{p^2+1}}, \qquad \int_{\mathbb{R}} \|F_1(p)\|_F^2\,dp = 2.$$

For the inverse transform $\mathcal{K}_1(s) = (2\pi)^{-1}\int_{\mathbb{R}} e^{ips}F_1(p)\,dp$, the explicit Bessel kernel gives $\|\mathcal{K}_1(s)\|_F^2 = G(|s|)/(2\pi^2)$ for $s \neq 0$. Fourier–Plancherel therefore yields

$$\frac{1}{\pi^2}\int_0^\infty G(t)\,dt = \int_{\mathbb{R}} \|\mathcal{K}_1(s)\|_F^2\,ds = \frac{1}{2\pi}\int_{\mathbb{R}} \|F_1(p)\|_F^2\,dp = \frac{1}{\pi},$$

so

$$\boxed{\int_0^\infty G(t)\,dt = \pi.} \tag{5}$$

Equations (4)–(5), together with $K_\nu(t) = O(e^{-t}t^{-1/2})$, establish $A(\mu) - H(\mu) \to 0$ as $\mu \to \infty$. Differentiating the convergent integrals,

$$H'(\mu) = \frac{1}{\pi} - \frac{1}{\pi^2}\int_\mu^\infty G(t)\,dt,$$
$$A'(\mu) = \frac{1}{\pi} - \frac{1}{\pi^2\mu}.$$

Since $G(t) = t^{-2} - W(t)$, setting $E = A - H$ gives

$$E'(\mu) = -\frac{1}{\pi^2}\int_\mu^\infty W(t)\,dt, \qquad E''(\mu) = \frac{W(\mu)}{\pi^2}. \tag{6}$$

The known vanishing of E at infinity and the exponential decay of W permit integration of (6) twice, giving the exact representation (1).

To prove its sign without numerical assumptions, use the standard representation

$$K_\nu(t) = \int_0^\infty e^{-t\cosh u}\cosh(\nu u)\,du, \qquad t > 0, \tag{7}$$

which gives $0 < K_0(t) < K_1(t)$. Also $(tK_1(t))' = -tK_0(t) < 0$, while $\lim_{t\downarrow 0} tK_1(t) = 1$. Thus $K_1(t) < 1/t$, and hence

$$0 < W(t) = \frac{2K_1(t)}{t} - K_0(t)^2 - K_1(t)^2 < \frac{2K_1(t)}{t}. \tag{8}$$

For $t \geq \mu$, (7) and $\cosh u \geq 1$ imply

$$K_1(t) \leq e^{-(t-\mu)}K_1(\mu). \tag{9}$$

Using $t^{-1} \leq \mu^{-1}$ in (8), insert (9) into (1) and evaluate $\int_0^\infty se^{-s}ds = 1$. This proves (2), including strictness. Equation (6) yields $E' < 0$ and $E'' > 0$.

Finally, the classical large-argument expansion for K_1 gives

$$W(t) = \sqrt{2\pi}\,e^{-t}t^{-3/2}\big(1 + O(t^{-1})\big) \qquad (t \to \infty), \tag{10}$$

because $K_0(t)^2 + K_1(t)^2 = O(e^{-2t}t^{-1})$. Substitute $t = \mu + s$ in (1), factor out $e^{-\mu}\mu^{-3/2}$, and integrate against se^{-s}. The uniform remainder estimate for $t \geq \mu$ gives relative error $O(\mu^{-1})$, proving (3). As (2) has the same leading equivalent, its ratio to the exact gap tends to 1. □

Further asymptotic consistency check. The same integral representation also permits two additional inverse-mass coefficients:

$$A(\mu) - H(\mu) = \frac{\sqrt{2\pi}}{\pi^2}\frac{e^{-\mu}}{\mu^{3/2}}\left(1 - \frac{21}{8\mu} + \frac{1185}{128\mu^2} + O(\mu^{-3})\right).$$

Indeed, the standard expansion $K_1(t) = \sqrt{\pi/(2t)}e^{-t}[1 + 3/(8t) - 15/(128t^2) + O(t^{-3})]$ implies $W(t) = \sqrt{2\pi}e^{-t}t^{-3/2}[1 + 3/(8t) - 15/(128t^2) + O(t^{-3})] + O(e^{-2t}/t)$. Setting $t = \mu + s$, expanding the remaining powers of $1 + s/\mu$ in the exponentially weighted integral, and using $\int_0^\infty s^{j+1}e^{-s}ds = (j+1)!$ gives the stated coefficients. The expansion is justified by the uniform large-t remainder and an integrable polynomial-times-e^{-s} envelope for $s \geq 0$ when μ is sufficiently large.

3. Computed checks

The following values are numerical checks of the exact formula, not ingredients of the proof. The gap is evaluated directly as the positive integral in (1), avoiding loss of precision through subtraction of two nearly equal large quantities.

μ	Exact-gap quadrature $A - H$	Global upper bound $2K_1/(\pi^2\mu)$	Gap divided by bound
0.5	$5.49950703804 \times 10^{-2}$	$6.71330299650 \times 10^{-1}$	0.08192
2	$5.19873489905 \times 10^{-3}$	$1.41713766968 \times 10^{-2}$	0.36685
5	$1.03260959912 \times 10^{-4}$	$1.63922008668 \times 10^{-4}$	0.62994
10	$2.92623963802 \times 10^{-7}$	$3.77903159964 \times 10^{-7}$	0.77434
20	$5.19584664189 \times 10^{-12}$	$5.96078396912 \times 10^{-12}$	0.87167

40	$4.00735359517 \times 10^{-21}$	$4.30469733616 \times 10^{-21}$	0.93093

In particular the two-term expression $A(\mu)$ is an *upper bound* for $H(\mu)$ at every positive μ, not merely an asymptotic approximation for large μ.

4. Scope

The continuum function H arises after taking the interval-length limit $L \to \infty$ with the scaling parameter $\mu > 0$ fixed. The subsequent $\mu \to \infty$ asymptotic is a second limit; no uniform joint-limit assertion is made. This calculation does not determine the logarithmic coefficient or finite part of the complete Wilson–Dirac quasifree relative entropy, whose finite-part convergence remains a separate problem. The leading entropy law with bounded remainder follows from the separate estimates assembled in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*.

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Uniform Square-Root-Mass Control of Wilson-Dirac Boundary Covariances

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Abstract

For the one-dimensional two-orbital Wilson–Dirac occupied-band projections, we establish a mass-uniform trace-class estimate for the difference of the cross-boundary blocks of the infinite-chain projections compressed to a finite interval. If the interval contains L cells, the mass is $m = \mu/L$, and $\log(1+m) \le 1/4$, then the boundary trace norm is at most $C\sqrt{\mu}$, for a universal constant C. Consequently, the quasifree interval entropies differ by at most $2C\sqrt{\mu}$. The proof uses a Fourier–Laplace/Hankel decomposition, retaining the finite-length square-root weight throughout the branch-point estimates. Combining this with the fixed-μ Hilbert–Schmidt continuum theorem shows that the exponent $1/2$ cannot be increased for a uniform estimate on the boundary trace norm. The entropy-difference limit is not determined by this bound.

1. Setting and theorem

Let P_m be the orthogonal Fourier-multiplier projection on $\ell^2(\mathbb{Z};\mathbb{C}^2)$ associated with the negative-energy band of

$$h_m(k) = \sin k\,\sigma_x + (1+m-\cos k)\sigma_z, \qquad P_m(k) = \tfrac{1}{2}(I - h_m(k)/E_m(k)),$$
$$E_m(k) = \sqrt{m^2 + 4(1+m)\sin^2(k/2)}.$$

The massless projector P_0 is defined almost everywhere. Let χ_L project onto any L consecutive two-orbital cells, and set

$$m = \mu/L > 0, \quad \kappa = \log(1+m), \quad Z_{L,\mu} = \chi_L(P_m - P_0)(I - \chi_L).$$

Theorem 1. There is a universal finite constant C such that, for every $L \ge 1$ and $\mu > 0$ satisfying $0 < \kappa \le 1/4$,

$$\boxed{\|Z_{L,\mu}\|_{S_1} \le C\sqrt{L\kappa} \le C\sqrt{\mu}.} \tag{1}$$

For the compressed covariance matrices $A_L = \chi_L P_m \chi_L$ and $B_L = \chi_L P_0 \chi_L$, on the interval subspace,

$$\boxed{|S(A_L) - S(B_L)| \le 2C\sqrt{\mu}.} \tag{2}$$

For every fixed $\mu > 0$, the condition on κ holds for all sufficiently large L. The constant does not depend on L or μ in that range.

2. Exact Fourier–Laplace representation

Write $a = e^\kappa = 1 + m$, and define, for $t > 0$,

$$h_\kappa(t) = \mathbf{1}_{t>\kappa}\frac{e^{-\kappa/2}}{\sqrt{2(\cosh t - \cosh\kappa)}}, \qquad h_0(t) = \frac{1}{2\sinh(t/2)},$$

$$\rho_x(t) = \sinh t\,[h_\kappa(t) - h_0(t)], \qquad \rho_z(t) = (e^\kappa - \cosh t)h_\kappa(t) - (1-\cosh t)h_0(t).$$

For $n \geq 1$, the positive-distance Fourier coefficient of $P_m - P_0$ is

$$T_n = -\frac{1}{2\pi} \int_0^\infty e^{-nt} [\rho_z(t)\sigma_z + i\rho_x(t)\sigma_x] dt, \qquad T_{-n} = T_n^*. \tag{3}$$

This follows by writing $E_m(k) = e^{\kappa/2}\sqrt{2(\cosh\kappa - \cos k)}$, inserting the Poisson kernel, and taking the cosine/sine Fourier coefficients. In particular the normalized cosine representation is

$$\frac{1}{\pi}\int_0^\pi \frac{\cos(nk)}{\sqrt{2(\cosh\kappa - \cos k)}} dk = \frac{1}{\pi}\int_\kappa^\infty \frac{e^{-nt}}{\sqrt{2(\cosh t - \cosh\kappa)}} dt.$$

All subsequent integrals involve the combination (3), in which the singular massless terms cancel; we do not assign a finite value to the individually divergent massless inverse-energy Fourier transform.

Derivation of the normalized transform. For $\kappa > 0$ set $g_\kappa(k) = [2(\cosh\kappa - \cos k)]^{-1/2}$. The Poisson kernel identity

$$\sum_{n\in\mathbb{Z}} e^{-|n|t} e^{ink} = \frac{\sinh t}{\cosh t - \cos k}$$

can be integrated against $[2(\cosh t - \cosh\kappa)]^{-1/2}\, dt$ over $t > \kappa$. The substitution $u = \sqrt{2(\cosh t - \cosh\kappa)}$ reduces the resulting integral to

$$\int_0^\infty \frac{du}{u^2/2 + \cosh\kappa - \cos k} = \pi g_\kappa(k).$$

Comparison of Fourier coefficients proves the displayed cosine transform. Multiplication by $1 + m - \cos k$ or $\sin k$ gives the z and x coefficients in (3). The limit $\kappa \downarrow 0$ is taken in these combinations, where the singular terms cancel.

The right and left exterior half-lines each give a matrix-valued Hankel block. Rank-one factorization of $e^{-t(i+j+1)}$, for $0 \leq i < L$, $j \geq 0$, has nuclear norm

$$W_L(t) = e^{-t}\frac{\sqrt{1 - e^{-2Lt}}}{1 - e^{-2t}}. \tag{4}$$

Taking the Schatten norm of the 2×2 orbital factor in (3), and accounting for both boundaries, yields the rigorous Bochner-integral estimate

$$\|Z_{L,\mu}\|_{S_1} \leq \frac{2}{\pi}\int_0^\infty W_L(t)(|\rho_x(t)| + |\rho_z(t)|) dt. \tag{5}$$

3. Strengthened finite-length kernel bound

The essential improvement is to keep the finite-L numerator of (4) instead of prematurely replacing it by one. Because $1 - e^{-2Lt} \leq 2Lt$ and $1 - e^{-2t} \geq (1 - e^{-2})t$ when $0 < t \leq 1$, there is a universal C_0 such that

$$\boxed{W_L(t) \leq C_0\sqrt{L/t} \quad (0 < t \leq 1), \qquad W_L(t) \leq C_0 e^{-t} \quad (t \geq 1).} \tag{6}$$

The bound need not be optimal for large t; its small-t behavior is what retains the $\sqrt{L\kappa}$ scale.

4. Four-region proof

Assume $0 < \kappa \leq 1/4$. All constants in this section are universal.

Region I: $0 < t < \kappa$**.** The massive density vanishes: $h_\kappa = 0$. Exactly,

$$\rho_x = -\cosh(t/2), \qquad \rho_z = \sinh(t/2).$$

Thus $|\rho_x| + |\rho_z| \le C_1$, and (6) gives

$$\int_0^\kappa W_L(|\rho_x| + |\rho_z|)dt \le C_2\sqrt{L}\int_0^\kappa t^{-1/2}dt \le C_3\sqrt{L\kappa}. \tag{7}$$

Region II: $\kappa < t < 2\kappa$**.** Elementary bounds on $\cosh t - \cosh\kappa$, the massive prefactor, and the massless density show

$$|\rho_x| + |\rho_z| \le C_4\left(1 + \frac{t+\kappa}{\sqrt{t^2-\kappa^2}}\right).$$

Set $t = \kappa u$ in (6). Then

$$\int_\kappa^{2\kappa} W_L(|\rho_x| + |\rho_z|)dt \le C_5\sqrt{L\kappa}\int_1^2 u^{-1/2}\left(1 + \frac{u+1}{\sqrt{u^2-1}}\right)du. \tag{8}$$

The final integral is finite: its singularity at $u = 1$ is only of order $(u-1)^{-1/2}$. Hence (8) is $O(\sqrt{L\kappa})$.

Region III: $2\kappa < t < 1$**.** Since $t \ge 2\kappa$, the denominator of h_κ is uniformly comparable with t; the exponential prefactor changes by $O(\kappa)$ and the denominator by $O(\kappa^2)$. Consequently,

$$|h_\kappa - h_0| \le C_6(\kappa/t + \kappa^2/t^3),$$

$$|\rho_x| \le C_7(\kappa + \kappa^2/t^2), \qquad |\rho_z| \le C_7\kappa/t.$$

Using (6), the region-III integral is bounded by

$$C_8\sqrt{L}\left[\kappa\int_{2\kappa}^1 t^{-1/2}dt + \kappa^2\int_{2\kappa}^1 t^{-5/2}dt + \kappa\int_{2\kappa}^1 t^{-3/2}dt\right] \le C_9\sqrt{L\kappa}. \tag{9}$$

All exponents have been kept explicit; the potentially difficult lower endpoint contributes only $O(\sqrt{L\kappa})$.

Region IV: $t \ge 1$**.** The inverse-energy denominators are separated from their singularities and

$$|\rho_x| + |\rho_z| \le C_{10}\kappa e^{t/2}.$$

With $W_L \le C_0 e^{-t}$, the final integral is $O(\kappa)$, which is $O(\sqrt{L\kappa})$ for $L \ge 1$ and $\kappa \le 1$.

Adding the estimates (7)–(9) and the fourth region in (5) proves the first assertion of Theorem 1. Finally $L\log(1+\mu/L) \le \mu$, proving the second inequality in (1).

5. Entropy continuity

For any infinite-space orthogonal projector P, let $X = \chi_L P(I-\chi_L)$ and $C = \chi_L P\chi_L$ on the interval. Then $XX^* = C(I-C)$. The singular-value entropy function

$$F(s) = h\left(\frac{1+\sqrt{1-4s^2}}{2}\right), \quad 0 \le s \le \tfrac{1}{2},$$

has derivative

$$F'(s) = \frac{4s}{\sqrt{1-4s^2}}\,\operatorname{artanh}\sqrt{1-4s^2}, \quad 0 \le F'(s) \le 2.$$

Mirsky's singular-value inequality and the scalar Lipschitz bound imply

$$|S(A_L) - S(B_L)| \le 2\|\chi_L(P_m - P_0)(I-\chi_L)\|_1.$$

Inserting (1) proves (2).

6. Optimality of the trace-norm exponent in the small-mass limit

The exponent in (1) cannot be raised uniformly. This conclusion also uses the separately established fixed-μ Hilbert–Schmidt continuum theorem; it does not follow from the four-region upper bound alone.

Indeed, write $\Delta_L = \chi_L(P_m - P_0)\chi_L$, $G_L = \chi_L(P_m - P_0)^2\chi_L$. Exact positive compression gives

$$\|Z_{L,\mu}\|_2^2 = \mathrm{Tr}(G_L - \Delta_L^2).$$

For fixed $\mu > 0$, the full-line Fourier trace and the continuum covariance-distance result in *Hilbert-Schmidt Scaling Limit for Wilson-Dirac Covariances* show

$$\lim_{L\to\infty} \|Z_{L,\mu}\|_2^2 = \Lambda(\mu) = \frac{\mu}{\pi} - H(\mu).$$

The continuum Hilbert–Schmidt scaling function is

$$H(\mu) = \frac{1}{\pi^2}\int_0^1 (1-s)\left[\mu^2 K_0(\mu s)^2 + \left(\frac{1}{s} - \mu K_1(\mu s)\right)^2\right] ds.$$

The classical small-argument Bessel expansions imply

$$K_0(z) = O(1 + |\log z|), \qquad K_1(z) - z^{-1} = O(z(1 + |\log z|)), \quad 0 < z \le 1,$$

so $H(\mu) = O(\mu^2(1 + |\log \mu|)^2) = o(\mu)$ as $\mu \downarrow 0$. Therefore

$$\lim_{\mu\downarrow 0} \frac{\Lambda(\mu)}{\mu} = \frac{1}{\pi}.$$

Since the nuclear norm dominates the Hilbert–Schmidt norm, we conclude

$$\boxed{\liminf_{\mu\downarrow 0} \liminf_{L\to\infty} \frac{\|Z_{L,\mu}\|_1}{\sqrt{\mu}} \ge \frac{1}{\sqrt{\pi}} > 0.} \tag{10}$$

Thus no estimate $\|Z_{L,\mu}\|_1 \le C\mu^{1/2+\epsilon}$, for any $\epsilon > 0$, can hold uniformly for all sufficiently large L and small μ in this sequence. **This optimality statement applies to the boundary trace norm, not to the entropy difference.**

7. Scope

The proof uses the specific Wilson–Dirac Bloch projectors, a finite interval of consecutive cells, and the mass scaling $m = \mu/L$. It does not establish uniformity for arbitrary mass sequences or other lattice Hamiltonians. It does not prove convergence of $S(A_L) - S(B_L)$ as $L \to \infty$ for a fixed positive μ. Nor does it establish a proposed finite entropy contribution, a uniform reference-modular spectral approximation, or the full relative-entropy logarithmic coefficient.

The stronger boundary convergence and entropy-difference limit are proved in *Cross-Boundary Trace-Class Estimates for Wilson–Dirac Quasifree States*. The complete leading logarithm is established in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*. These are separate arguments. Theorem 1 of *Finite Part of Wilson–Dirac Relative Entropy* combines the boundary limit with modular and nonlinear convergence to obtain the complete fixed-positive-scaled-mass finite part and an $O_\mu(1/\log L)$ remainder. The bound proved in this article alone does not establish that limit.

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Exact Longitudinal Jacobi Moment and Wilson Majorant Mass for a Finite Interval

Kirtus Justus (Theoretical Physicist) • 31 October 2026
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Abstract

For the occupied-band Wilson–Dirac projection compressed from the infinite lattice to a finite interval of two-orbital cells, we derive closed elliptic expressions for the first Jacobi moment of the longitudinal perturbation and for the total mass of its positive spectral majorant. At fixed scaled mass, the Jacobi moment has an explicitly determined logarithmic coefficient and finite constant. These statements concern the longitudinal operator against the *Jacobi generator*, not against the exact finite reference-modular discrepancy. They do not settle the leading coefficient or the finite part of the complete Wilson relative entropy.

1. Definitions and scalar Fourier normalization

Let $N = 2L$ and $m = \mu/L$ with $\mu > 0$ fixed, and put

$$r = \frac{m}{m+2} = \frac{\mu}{N+\mu} \in (0,1), \qquad z = 1 - r^2.$$

The reference Jacobi operator on $\mathbb{C}^N$ is the real symmetric matrix

$$(J_N)_{j-1,j} = (J_N)_{j,j-1} = \frac{j(N-j)}{N}, \qquad 1 \le j < N,$$

with zero diagonal. The longitudinal Wilson perturbation S_N is the compression, in the fixed interval-preserving scalar basis, of the full-line operator $S = G(I - 2P_0)$, with P_0 the half-filled occupied-band projection and G the positive Wilson majorant. The full-line Fourier multiplier is scalar Toeplitz after transformation; its odd scalar displacement coefficients are

$$s_d(r) = -\frac{1}{\pi}\int_0^{\pi/2} \left(1 - \frac{\cos\theta}{\sqrt{\cos^2\theta + r^2\sin^2\theta}}\right)\cos(d\theta)\,d\theta, \qquad d \text{ odd},$$

while $s_d = 0$ for even displacement, including $s_0 = 0$. This identity follows directly from the full-line Wilson symbol, its positive-majorant factorization and the fixed cellwise scalarization. No finite-open-chain projector is introduced. Let $\mathbf{K}(z)$ and $\mathbf{E}(z)$ denote complete elliptic integrals with **parameter**, not modulus, z.

2. Exact first Jacobi moment

Theorem 1. For every even $N \ge 2$ and $\mu > 0$,

$$\boxed{\operatorname{Tr}(S_N J_N) = \frac{N^2-1}{3\pi}\left[\frac{\mathbf{E}(1-r^2) - r^2\mathbf{K}(1-r^2)}{1-r^2} - 1\right].} \tag{1}$$

At fixed $\mu > 0$, as $N \to \infty$ along even integers,

$$\boxed{\operatorname{Tr}(S_N J_N) = -\frac{\mu^2}{6\pi}\log N + \frac{\mu^2}{4\pi} - \frac{\mu^2}{6\pi}\log\frac{4}{\mu} + O_\mu\Big(\frac{\log N}{N}\Big).} \tag{2}$$

Proof. As J_N has first off-diagonals only, and S_N has Toeplitz first off-diagonal $s_1(r)$,

$$\operatorname{Tr}(S_N J_N) = 2s_1(r) \sum_{j=1}^{N-1} \frac{j(N-j)}{N} = \frac{N^2-1}{3} s_1(r).$$

The first coefficient of S_N is

$$s_1(r) = -\frac{1}{\pi}\left[1 - \int_0^{\pi/2} \frac{\cos^2\theta}{\sqrt{1-(1-r^2)\sin^2\theta}}\, d\theta\right].$$

The elementary identity

$$\int_0^{\pi/2} \frac{\cos^2\theta}{\sqrt{1-z\sin^2\theta}}\, d\theta = \mathbf{K}(z) - \frac{\mathbf{K}(z)-\mathbf{E}(z)}{z} = \frac{\mathbf{E}(z)-(1-z)\mathbf{K}(z)}{z}$$

gives (1). As $r \downarrow 0$, standard complete elliptic expansions yield

$$\mathbf{K}(1-r^2) = \log\frac{4}{r} + O(r^2\log(1/r)), \quad \mathbf{E}(1-r^2) = 1 + \frac{r^2}{2}\left(\log\frac{4}{r} - \frac{1}{2}\right) + O(r^4\log(1/r)).$$

Consequently,

$$s_1(r) = \frac{r^2}{\pi}\left(\frac{3}{4} - \frac{1}{2}\log\frac{4}{r}\right) + O(r^4\log(1/r)).$$

Substitute $r = \mu/(N+\mu)$ into the finite identity. Since $N^2r^2 = \mu^2 + O_\mu(N^{-1})$ and $\log(4/r) = \log N + \log(4/\mu) + O_\mu(N^{-1})$, equation (2) follows with the stated remainder. □

3. Exact mass of the positive Wilson majorant

The full-line positive majorant has the orbital-scalar Bloch multiplier

$$G(k) = \frac{1}{2} g_r(k) I_2, \qquad g_r(k) = 1 - \frac{|\sin(k/2)|}{\sqrt{\sin^2(k/2) + r^2\cos^2(k/2)}}.$$

Under the interval-preserving scalarization, its diagonal average is unchanged.

Theorem 2. For every even $N \geq 2$ and $\mu > 0$,

$$\boxed{\operatorname{Tr} G_N = N\left[\frac{1}{2} - \frac{\arccos r}{\pi\sqrt{1-r^2}}\right].} \tag{3}$$

For fixed $\mu > 0$,

$$\boxed{\operatorname{Tr} G_N = \frac{\mu}{\pi} - \frac{\mu^2}{N}\left(\frac{1}{\pi} + \frac{1}{4}\right) + O_\mu(N^{-2}).} \tag{4}$$

Proof. The trace is N times the zeroth scalar Fourier coefficient of G. By symmetry,

$$\frac{1}{N}\operatorname{Tr} G_N = \frac{1}{\pi}\int_0^{\pi/2}\left(1 - \frac{\sin\theta}{\sqrt{\sin^2\theta + r^2\cos^2\theta}}\right) d\theta.$$

The first term integrates to $1/2$. Substitute $u = \cos\theta$ in the second term to obtain

$$\int_0^1 \frac{du}{\sqrt{1-(1-r^2)u^2}} = \frac{\arcsin\sqrt{1-r^2}}{\sqrt{1-r^2}} = \frac{\arccos r}{\sqrt{1-r^2}}.$$

This proves (3). Taylor-expanding $\arccos r/\sqrt{1-r^2} = \pi/2 - r + (\pi/4)r^2 + O(r^3)$, and substituting $r = \mu/(N+\mu)$, proves (4). □

Consequence. The boundary-corrected modular estimate can retain the actual finite majorant mass:

$$2|\operatorname{Tr}(C_N E_N)| \le \sqrt{2\operatorname{Tr} G_N}\,\sqrt{\operatorname{Tr}(W_N E_N^2)}.$$

This sharpens the numerical prefactor in the bound obtained from μ/π but does not by itself improve the $O_\mu(\sqrt{\log N})$ bound obtained by this method. The separate weighted theorem in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* gives a stronger estimate. The cross-boundary Hilbert–Schmidt leakage is bounded above by $\frac{1}{2}\operatorname{Tr} G_N$.

4. Scope of the modular application

The exact moment identifies the first Jacobi moment of the longitudinal perturbation, not the modular trace $\operatorname{Tr}(S_N E_N)$. An approximation to E_N that is valid only for central Jacobi modes cannot be substituted into the full trace without an appropriate uniform, weighted error estimate. The reference-modular estimates in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* establish the complete leading law with bounded remainder. Convergence to a finite part still requires further estimates; the moment identity here does not provide them.

The cross-boundary leakage of the positive majorant has an exact fixed-scaled-mass limit, given in *Continuum Limit of the Wilson Majorant Boundary Defect*. The majorant mass and the boundary-leakage norm are different quantities.

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Kirtus Justus directed the mathematical research and established the publication scope. ChatGPT (OpenAI) derived and checked the elliptic moment and majorant-mass identities and prepared the manuscript. Claude (Anthropic), Gemini (Google), Grok (SpaceXAI), and DeepSeek (High-Flyer) are acknowledged for documented broader participation only, without attributing independent verification of these results to them. No external peer review, experimental validation, or formal proof-assistant verification is claimed.

Boundary-Corrected Longitudinal Modular Reduction for a Wilson–Dirac Interval

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Abstract

For a finite interval cut from the infinite Wilson–Dirac occupied-band projection, the longitudinal perturbation can be rewritten exactly as a positive-majorant-weighted reference term plus a cross-boundary correction. The correction admits an explicit Hilbert–Schmidt bound controlled by the occupation-weighted modular quadratic form. As a consequence, the leading-order relative-entropy reduction can be equivalently formulated using the positive majorant and the reference occupation operator. This argument reformulates the modular estimate. The separate whole-line argument in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* controls that estimate and establishes the leading entropy law.

1. Operator setting

Let $\mathcal{H} = \ell^2(\mathbb{Z}) \otimes \mathbb{C}^2$ and let χ be the orthogonal projection onto L consecutive two-orbital cells, $N = 2L$. Operators on $\operatorname{ran}\chi$ are implicitly restricted to that space. All full-line operators are expressed after one common, interval-preserving cellwise rotation and staggering.

Let P_0 be the massless occupied-band projection and G the positive Wilson spectral majorant. For fixed $\mu > 0$ and $m = \mu/L$, the full-line projector calculation gives

$$P_0^2 = P_0 = P_0^*, \qquad G = G^*, \qquad 0 \le G \le \tfrac{1}{2} I, \qquad [G, P_0] = 0,$$
$$S = G(I - 2P_0), \qquad \operatorname{Tr}(\chi G \chi) \le \frac{\mu}{\pi}. \tag{1}$$

Define

$$B_N = \chi P_0 \chi, \qquad G_N = \chi G \chi, \qquad S_N = \chi S \chi, \qquad W_N = B_N(I - B_N). \tag{2}$$

The compressed reference satisfies $0 < B_N < I$. Its commuting Jacobi operator J_N has off-diagonal entries $j(N-j)/N$. Put

$$K_N = \log(I - B_N) - \log B_N, \quad E_N = K_N + N\eta(J_N/N), \tag{3}$$

with the half-filled reference scaling function η from the free-fermion Jacobi model. In particular, $E_N = E_N^*$ and $[E_N, B_N] = 0$. The occupation-weighted estimate is

$$\operatorname{Tr}(W_N E_N^2) \le C(1 + \log N) \quad (N \ge 2), \tag{4}$$

with a constant independent of N. We explicitly retain (1) and (4) as input hypotheses to the new theorem.

2. Exact boundary-corrected decomposition

Theorem 1. Let $Q = I - \chi$ and set

$$C_N = \chi G Q P_0 \chi.$$

Then for every finite interval,

$$\boxed{S_N = G_N - G_N B_N - B_N G_N - C_N - C_N^*.} \tag{5}$$

Consequently,

$$\boxed{\operatorname{Tr}(S_N E_N) = \operatorname{Tr}\big[G_N (I - 2B_N) E_N\big] - 2 \operatorname{Re} \operatorname{Tr}(C_N E_N).} \tag{6}$$

Proof. From $S = G(I - 2P_0)$ and $[G, P_0] = 0$,

$$S_N = G_N - \chi (G P_0 + P_0 G) \chi.$$

Insert $I = \chi + Q$ between the two full-line operators in each product:

$$\begin{aligned} \chi G P_0 \chi &= G_N B_N + \chi G Q P_0 \chi, \\ \chi P_0 G \chi &= B_N G_N + \chi P_0 Q G \chi. \end{aligned}$$

Because G, P_0 are self-adjoint, the last term is C_N^*. This proves (5), without assuming that G_N commutes with B_N. Multiply by E_N, trace, use $[E_N, B_N] = 0$ and cyclicity, and combine the conjugate boundary traces to obtain (6). □

3. Sublogarithmic boundary control

Theorem 2. Under the hypotheses (1)–(4),

$$\boxed{2\,|\operatorname{Tr}(C_N E_N)| \le \sqrt{\frac{2\mu}{\pi}}\,\big[\operatorname{Tr}(W_N E_N^2)\big]^{1/2} = O_\mu(\sqrt{\log N}).} \tag{7}$$

Proof. The two factors in $C_N = (QG\chi)^*(QP_0\chi)$ are Hilbert–Schmidt after restriction to the finite interval. Because $0 \le Q \le I$ and $G^2 \le G/2$,

$$\begin{aligned} \|QG\chi\|_{\mathrm{HS}}^2 &= \operatorname{Tr}(\chi G Q G \chi) \\ &\le \operatorname{Tr}(\chi G^2 \chi) \le \tfrac{1}{2} \operatorname{Tr} G_N \le \frac{\mu}{2\pi}. \end{aligned} \tag{8}$$

The full-line projection identity gives

$$\chi P_0 Q P_0 \chi = \chi P_0 \chi - (\chi P_0 \chi)^2 = W_N,$$

and hence

$$\|Q P_0 \chi E_N\|_{\mathrm{HS}}^2 = \operatorname{Tr}(E_N W_N E_N) = \operatorname{Tr}(W_N E_N^2). \tag{9}$$

Cauchy–Schwarz for the Hilbert–Schmidt inner product, (8), (9), and (4) prove the assertion. □

The exact Hilbert–Schmidt factor is the boundary leakage $\Lambda_N = \operatorname{Tr}(\chi G Q G \chi) = \operatorname{Tr}(\chi G^2 \chi) - \operatorname{Tr}(G_N^2)$. Consequently, the proof also gives

$$2|\operatorname{Tr}(C_N E_N)| \le 2\sqrt{\Lambda_N}\,[\operatorname{Tr}(W_N E_N^2)]^{1/2}.$$

For fixed $\mu > 0$, the companion theorem *Continuum Limit of the Wilson Majorant Boundary Defect* establishes $\Lambda_N \to \Phi(\mu) > 0$. This identifies the asymptotic Hilbert–Schmidt prefactor without changing the $O_\mu(\sqrt{\log N})$ growth order.

4. Equivalent leading-order obstruction

The established four-branch entropy decomposition and its auxiliary estimates give, for fixed $\mu > 0$,

$$\mathcal{D}_L(\mu) = \frac{\mu^2}{6}\log L + \operatorname{Tr}(S_N E_N) + O_\mu(\sqrt{\log L}), \qquad N = 2L. \tag{10}$$

Corollary 3. Under the same hypotheses,

$$\boxed{\mathcal{D}_L(\mu) = \frac{\mu^2}{6}\log L + \operatorname{Tr}\big[G_N(I - 2B_N)E_N\big] + O_\mu(\sqrt{\log L}).} \tag{11}$$

In a common orthonormal eigenbasis $B_N v_j = b_j v_j$, $J_N v_j = \lambda_j v_j$, with $E_N v_j = e_j v_j$ and $q_j = \langle v_j, G_N v_j\rangle \ge 0$, the remaining bulk trace is exactly

$$\boxed{\operatorname{Tr}\big[G_N(I - 2B_N)E_N\big] = \sum_{j=1}^{N} q_j(1 - 2b_j)e_j.} \tag{12}$$

Thus the original condition $\operatorname{Tr}(S_N E_N) = o(\log N)$ is **equivalent** to the signed bulk condition

$$\sum_j q_j(1 - 2b_j)e_j = o(\log N). \tag{13}$$

A sufficient noncancellation criterion is

$$\boxed{\sum_j q_j|1 - 2b_j|\,|e_j| = o(\log N).} \tag{14}$$

It is no stronger than the stronger sufficient bound $\sum_j q_j|e_j| = o(\log N)$, since $|1 - 2b_j| \le 1$; the extra occupation factor suppresses contributions from reference levels near half-filling. The separate weighted theorem in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* proves the stronger bound $\sum_j q_j|e_j| = O_\mu(1)$, and hence both conditions at fixed positive scaled mass. Numerical tests alone would not prove either condition.

5. Scope and limitations

Equations (5) and (6) are exact at finite interval length. Equation (7) is a proved consequence of the explicitly identified Wilson-majorant and weighted-modular inputs. Equation (11) is a reduction with a rigorously bounded error; identifying its leading coefficient additionally uses the weighted theorem cited next. The bulk trace in (13) is controlled by the separate weighted theorem in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*. A finite entropy part and an inverse-logarithmic convergence rate do not follow from this reduction alone. For the specified interval restrictions with fixed positive scaled mass, Theorem 1 of *Finite Part of Wilson–Dirac Relative Entropy* supplies that limit and rate.

The interval is a compression of the *infinite-chain occupied-band projection*. None of the statements concerns ground-state projectors of a finite open Wilson chain unless additional arguments establish that identification. Operators on the full line are never replaced by products of their compressions without the boundary terms displayed above.

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- K. Justus, "Continuum Limit of the Wilson Majorant Boundary Defect" (2026), for the fixed-scaled-mass limit of the Hilbert–Schmidt factor.

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Kirtus Justus directed the research program and supplied its mathematical starting point. ChatGPT (OpenAI) developed the displayed boundary decomposition, its Hilbert–Schmidt estimate, and the accompanying computational cross-checks; these derivations have not undergone independent external peer review. Claude (Anthropic), Gemini (Google), Grok (SpaceXAI), and DeepSeek (High-Flyer) are acknowledged as participants in the broader research effort without assigning them unverified contributions to this argument. No institutional affiliation, endorsement, or peer-review status is implied.

Continuum Limit of the Wilson Majorant Boundary Defect

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Abstract

For the positive Wilson majorant compressed to an even-length scalar interval in the fixed-scaled-mass regime, we prove existence of the limiting cross-boundary Hilbert–Schmidt leakage and express it by an absolutely convergent one-dimensional integral. We identify a positive finite limit, establish strict monotonicity and strict concavity of the limiting function, and determine its small-scaled-mass expansion. The result concerns the majorant alone; it does not imply a uniform estimate for the finite reference-modular discrepancy or the complete Wilson relative entropy.

Definitions

Fix $\mu > 0$, let $N \to \infty$ through even integers, and set $r_N = \mu/(N+\mu)$. On the cell Brillouin zone let the orbital-scalar Wilson majorant have symbol

$$G_r(k) = \tfrac{1}{2} g_r(k), \qquad g_r(k) = 1 - \frac{|\sin(k/2)|}{\sqrt{\sin^2(k/2) + r^2\cos^2(k/2)}}.$$

Let χ_N project onto the N-site scalar interval, with the cell-preserving scalarization for which the nonzero off-diagonal coefficients occur at even scalar separations. For $d \in \mathbb{Z}$, let

$$c_d(r) = \frac{1}{2\pi}\int_{-\pi}^{\pi} G_r(k) e^{-idk}\, dk = \frac{1}{2\pi}\int_0^{\pi} g_r(k)\cos(dk)\, dk.$$

The coefficient at scalar separation $2d$ has magnitude $|c_d(r)|$; possible staggering phases do not affect its squared modulus. Define

$$f(t) = 1 - \frac{t}{\sqrt{1+t^2}}, \qquad F(x) = \int_0^\infty f(t)\cos(2xt)\, dt \quad (x \geq 0).$$

The function f is nonnegative and integrable, with $\int_0^\infty f(t)\, dt = 1$; hence F is continuous and $F(0) = 1$.

Theorem: exact limiting leakage

Theorem 1. For every fixed $\mu > 0$,

$$\boxed{\lim_{\substack{N\to\infty \\ N \text{ even}}} \|(I - \chi_N) G_{r_N} \chi_N\|_{\mathrm{HS}}^2 = \Phi(\mu) := \frac{2}{\pi^2}\int_0^\infty \min\{\mu, 2x\}\, F(x)^2\, dx.}$$

The integral converges absolutely and $0 < \Phi(\mu) < \infty$.

Proof. For $r \in (0,1)$, set $h_r(k) = g_r(k)$ on $[0,\pi]$. Under $t = \tan(k/2)/r$, the Fourier coefficient becomes exactly

$$c_d(r) = \frac{r}{\pi} \int_0^\infty \frac{f(t)}{1+r^2t^2} \cos\bigl(2d \arctan(rt)\bigr)\, dt. \tag{1}$$

For each fixed $A < \infty$, dominated convergence, with the integral split at a fixed large t-cutoff, shows

$$\sup_{d\ge 0:\, rd\le A} \left| \frac{\pi c_d(r)}{r} - F(rd) \right| \longrightarrow 0 \qquad (r \downarrow 0). \tag{2}$$

Indeed the phase difference tends uniformly to zero for $rd \le A$ on compact t-ranges, while $f \in L^1$ controls the tails.

A global Fourier coefficient bound supplies the necessary domination. Direct differentiation on $(0,\pi)$, writing $s = \sin(k/2)$, $c = \cos(k/2)$ and $D^2 = s^2 + r^2c^2$, yields

$$h_r'(k) = -\frac{r^2 c}{2D^3}, \quad h_r''(k) = \frac{r^2 s}{4D^3} + \frac{3r^2(1-r^2)sc^2}{4D^5} \ge 0.$$

Thus $h_r'(0) = -1/(2r)$, $h_r'(\pi) = 0$, and $\int_0^\pi h_r''(k)\, dk = 1/(2r)$. Two integrations by parts for $d \ge 1$ give

$$|c_d(r)| \le \frac{1}{2\pi r d^2}. \tag{3}$$

On the other hand, (1) and $\int f = 1$ imply $|c_d(r)| \le r/\pi$ for every d. Together,

$$\boxed{\left| \frac{\pi c_d(r)}{r} \right| \le \min\left\{1, \frac{1}{2(rd)^2}\right\}, \qquad d \ge 1.} \tag{4}$$

The exact Fourier boundary identity, including the two signs of each nonzero scalar displacement, reads

$$L_N := \|(I - \chi_N) G_{r_N} \chi_N\|_{\mathrm{HS}}^2 = 2 \sum_{d=1}^\infty \min(N, 2d) |c_d(r_N)|^2. \tag{5}$$

Rewriting (5) as a mesh sum gives

$$L_N = \frac{2}{\pi^2} \sum_{d=1}^\infty r_N \min(N r_N, 2d r_N) \left| \frac{\pi c_d(r_N)}{r_N} \right|^2.$$

Since $N r_N \to \mu$, equation (2) provides uniform convergence of the summands on bounded intervals in the mesh variable $x = d r_N$. The global bound (4) dominates the mesh tails by an integrable multiple of $\min\{\mu+1, 2x\} \min\{1, (2x^2)^{-1}\}^2$. Consequently the mesh sums converge to the stated integral. Positivity follows from $F(0) = 1$, continuity, and $\mu > 0$; finiteness follows from (4), which also yields $|F(x)| \le \min\{1, (2x^2)^{-1}\}$ for $x > 0$. □

Structural properties and small-mass expansion

Theorem 2. The limiting function Φ is increasing and concave on $(0,\infty)$, with

$$\Phi'(\mu) = \frac{2}{\pi^2}\int_{\mu/2}^{\infty} F(x)^2\,dx > 0, \qquad \Phi''(\mu) = -\frac{1}{\pi^2}F(\mu/2)^2 \le 0.$$

Furthermore,

$$\boxed{\Phi(\mu) = \left(\frac{1}{\pi} - \frac{1}{4}\right)\mu - \frac{\mu^2}{2\pi^2} + o(\mu^2) \quad (\mu \downarrow 0).}$$

Proof. Differentiating the defining integral at the moving interface $x = \mu/2$ yields the two derivative formulas; the interface contributions cancel. To see strict positivity of $F(x)$ for $x > 0$, integrate by parts to write $F(x) = (2x)^{-1}\int_0^\infty h(t)\sin(2xt)\,dt$, with $h(t) = (1+t^2)^{-3/2}$ positive and strictly decreasing. Pairing successive positive and negative half-periods of the sine wave gives a strictly positive contribution from each pair. Hence $F(x) > 0$ for all $x > 0$, so $\Phi'(\mu) > 0$ and $\Phi''(\mu) < 0$ for all $\mu > 0$. For the small-μ expansion, write

$$\Phi(\mu) = \frac{2\mu}{\pi^2}\int_0^\infty F(x)^2\,dx - \frac{2}{\pi^2}\int_0^{\mu/2}(\mu - 2x)F(x)^2\,dx.$$

Cosine-transform Plancherel, accounting for the frequency $2x$, gives

$$\int_0^\infty F(x)^2\,dx = \frac{\pi}{4}\int_0^\infty f(t)^2\,dt = \frac{\pi}{4}\left(2 - \frac{\pi}{2}\right).$$

For the final equality put $t = \sinh u$, followed by $y = e^{-u}$, to obtain $\int f^2 dt = 2\int_0^1 y^2/(1+y^2)\,dy = 2 - \pi/2$. Since $F(x)^2 \to 1$ as $x \downarrow 0$, the final integral is $\mu^2/4 + o(\mu^2)$. Substitution proves the expansion. □

The identity $F(x) = (2x)^{-1}\int_0^\infty \sin(2xt)(1+t^2)^{-3/2}\,dt$ for $x > 0$ follows by integration by parts and can be used for stable numerical quadrature. The half-period comparison in Theorem 2 establishes strict concavity for every $\mu > 0$.

Application and limitation

If $C_N = \chi_N G_{r_N}(I - \chi_N)P_0\chi_N$, then the cross-boundary Cauchy–Schwarz argument implies

$$2|\operatorname{Tr}(C_N E_N)| \le 2\sqrt{L_N}\,\sqrt{\operatorname{Tr}[B_N(I - B_N)E_N^2]}.$$

Theorem 1 supplies $\sqrt{L_N} = \sqrt{\Phi(\mu)} + o(1)$ at each fixed $\mu > 0$. Combined with the occupation-weighted quadratic modular estimate, this retains an $O_\mu(\sqrt{\log N})$ boundary bound with an asymptotically identified Hilbert–Schmidt factor. It neither proves the required weighted bulk trace is sublogarithmic nor controls the full modular operator norm. The leading coefficient is established by the separate argument in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*; neither the entropy finite part nor an all-mode operator-norm bound follows from this boundary-defect limit.

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Conformal Invariance and Sharp Wilson Majorant Boundary Bounds

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Abstract

For the scalarized positive Wilson majorant, we establish an exact mass-independent weighted Fourier identity and a uniform bound on finite-interval Hilbert–Schmidt boundary leakage. The proof identifies the majorant symbol at each positive mass parameter with a disk-automorphic composition of one reference symbol and applies conformal invariance of the boundary homogeneous half-Sobolev seminorm. We derive the exact deficit from the saturated boundary energy and its fixed-mass finite-interval asymptotic. For the fixed-scaled-mass continuum leakage function, we determine its exact global supremum and its two leading inverse-power corrections at large scaled mass. The resulting uniform estimate improves the prefactor available in the reference-modular boundary-term bound, but does not settle the longitudinal reference-modular trace or the complete relative-entropy asymptotic.

1. Conventions

For a parameter $r > 0$, define the real, even 2π-periodic scalar Wilson-majorant symbol

$$G_r(k) = \frac{1}{2}\left(1 - \frac{|\sin(k/2)|}{\sqrt{\sin^2(k/2) + r^2\cos^2(k/2)}}\right), \qquad -\pi \le k \le \pi,$$

with Fourier coefficients

$$c_d(r) = \frac{1}{2\pi}\int_{-\pi}^{\pi} G_r(k) e^{-idk}\, dk.$$

In the cell-preserving scalarization of the two-orbital lattice, these coefficients occur at scalar displacements $2d$. For a scalar interval of even length N and projection χ_N, the exact full-line Fourier boundary identity is

$$\mathcal{L}_N(r) := \|(I - \chi_N)G_r\chi_N\|_{\mathrm{HS}}^2 = 2\sum_{d=1}^{\infty} \min(N, 2d)|c_d(r)|^2. \tag{1}$$

This identity counts the two directions across the interval boundary and uses the Fourier coefficient normalization above. It is not a statement about an arbitrary two-orbital matrix-valued symbol under a different scalarization.

2. Universal weighted Fourier identity

Theorem 1. For every $r > 0$,

$$\boxed{4\sum_{d=1}^{\infty} d\,|c_d(r)|^2 = \frac{1}{2\pi^2}.} \tag{2}$$

Consequently, for every admissible interval length N,

$$\boxed{0 < \mathcal{L}_N(r) < \frac{1}{2\pi^2},} \tag{3}$$

and its exact positive deficit is

$$\boxed{\frac{1}{2\pi^2} - \mathcal{L}_N(r) = 2 \sum_{d>N/2} (2d - N)|c_d(r)|^2.} \tag{4}$$

Proof. Put $a = (1-r)/(1+r) \in (-1,1)$, and let

$$M_a(z) = \frac{z-a}{1-az}, \qquad |z| = 1.$$

Writing $M_a(e^{ik}) = e^{i\psi_r(k)}$, the half-angle identity gives

$$\tan\frac{\psi_r(k)}{2} = \frac{1}{r}\tan\frac{k}{2}.$$

The formula defining G_r therefore gives the pointwise boundary identity

$$G_r(e^{ik}) = G_1(M_a(e^{ik})), \qquad G_1(k) = \tfrac{1}{2}(1 - |\sin(k/2)|). \tag{5}$$

For a real boundary function u with Fourier coefficients $\widehat{u}_n$, define the homogeneous half-Sobolev energy

$$\mathcal{E}(u) = \sum_{n\in\mathbb{Z}} |n|\,|\widehat{u}_n|^2 = \frac{1}{4\pi^2}\int_0^{2\pi}\!\!\int_0^{2\pi} \frac{|u(e^{i\theta}) - u(e^{i\varphi})|^2}{|e^{i\theta} - e^{i\varphi}|^2}\, d\theta\, d\varphi. \tag{6}$$

The identity holds first for trigonometric polynomials, by orthogonality and the Dirichlet-kernel identity, and then by approximation in the homogeneous $H^{1/2}$ norm. Its normalization can be checked on $u(e^{ik}) = e^{ik}$. A disk automorphism satisfies

$$|M_a(z) - M_a(w)|^2 = |M_a'(z)|\,|M_a'(w)|\,|z-w|^2, \quad |z| = |w| = 1,$$

and its boundary angular Jacobian equals $|M_a'(z)|$. Changing both variables in (6) proves $\mathcal{E}(u \circ M_a) = \mathcal{E}(u)$. This also follows from conformal invariance of the Dirichlet energy of harmonic extensions. In particular, (5) implies $\mathcal{E}(G_r) = \mathcal{E}(G_1)$.

For $d \geq 1$, elementary integration yields

$$c_d(1) = -\frac{1}{2\pi}\int_0^{\pi} \sin(k/2)\cos(dk)\, dk = \frac{1}{\pi(4d^2-1)}. \tag{7}$$

Since G_r is real and even, $\mathcal{E}(G_r) = 2\sum_{d\geq 1} d|c_d(r)|^2$. The partial sum at $r = 1$ telescopes:

$$4\sum_{d=1}^{D} d|c_d(1)|^2 = \frac{4}{\pi^2}\sum_{d=1}^{D}\frac{d}{(4d^2-1)^2} = \frac{1}{2\pi^2}\left(1 - \frac{1}{(2D+1)^2}\right). \tag{8}$$

Letting $D \to \infty$ and using invariance gives (2). Comparing (1) with the saturated sum (2) proves (3) and (4); strictness follows because the boundary symbol is not a trigonometric polynomial and hence has nonzero Fourier coefficients at arbitrarily large indices. □

3. Fixed-parameter finite-interval deficit

Theorem 2. For each fixed $r > 0$, as $N \to \infty$ through even integers,

$$\boxed{\mathcal{L}_N(r) = \frac{1}{2\pi^2} - \frac{1}{6\pi^2 r^2 N^2} + o_r(N^{-2}).} \tag{9}$$

Proof. Let $h_r(k) = 2G_r(k)$ for $0 \le k \le \pi$. Its one-sided endpoint derivative is $h_r'(0) = -1/(2r)$, while $h_r'(\pi) = 0$, and $h_r'' \in L^1(0,\pi)$ at every fixed $r > 0$. Twice integrating the cosine coefficient by parts, then applying the Riemann–Lebesgue lemma, gives

$$\lim_{d\to\infty} d^2 c_d(r) = \frac{1}{4\pi r}. \tag{10}$$

Using (4) with $A = N/2$, the asymptotic (10) is uniform in the tail $d > A$ as $A \to \infty$, whence

$$\begin{aligned}
\frac{1}{2\pi^2} - \mathcal{L}_N(r) &= 4\sum_{d>A} (d-A)|c_d(r)|^2 \\
&= \frac{1+o_r(1)}{4\pi^2 r^2} \sum_{d>A} \frac{d-A}{d^4} \\
&= \frac{1+o_r(1)}{4\pi^2 r^2} \left(\frac{1}{6A^2} + o(A^{-2}) \right) = \frac{1}{6\pi^2 r^2 N^2} + o_r(N^{-2}).
\end{aligned}$$

The asymptotic is for **fixed** r. It must not be used as a uniform approximation when r depends on N. □

4. Exact continuum saturation and its large-mass expansion

For $\mu > 0$, set $r_N = \mu/(N+\mu)$. Define

$$f(t) = 1 - \frac{t}{\sqrt{1+t^2}}, \qquad F(x) = \int_0^\infty f(t)\cos(2xt)\,dt.$$

The continuum boundary-leakage function is

$$\Phi(\mu) = \frac{2}{\pi^2} \int_0^\infty \min(\mu, 2x) F(x)^2\,dx. \tag{11}$$

This expression is the fixed-scaled-mass limit of $\mathcal{L}_N(r_N)$; its derivation from scaled Fourier coefficients is independent of the conclusions below.

Theorem 3. The following weighted transform integral is exact:

$$\boxed{\int_0^\infty xF(x)^2\,dx = \frac{1}{8}.} \tag{12}$$

Consequently,

$$\boxed{\lim_{\mu\to\infty} \Phi(\mu) = \sup_{\mu>0} \Phi(\mu) = \frac{1}{2\pi^2}.} \tag{13}$$

Moreover, as $\mu \to \infty$,

$$\boxed{\Phi(\mu) = \frac{1}{2\pi^2} - \frac{1}{6\pi^2\mu^2} - \frac{3}{10\pi^2\mu^4} + O(\mu^{-6}).} \tag{14}$$

Proof. We first give a Fourier-analytic derivation of (12) independent of the Möbius argument. Set $h(t) = (1+t^2)^{-3/2}$ and $H(k) = \int_0^\infty h(t)\sin(kt)\,dt$. Integration by parts yields $F(x) = H(2x)/(2x)$ for $x > 0$. A Laplace regularization of the positive quadratic integral gives

$$\int_0^\infty \frac{H(k)^2}{k}\,dk = \frac{1}{2}\int_0^\infty \int_0^\infty h(t)h(s)\log\frac{t+s}{|t-s|}\,dt\,ds. \tag{15}$$

Indeed, inserting $e^{-\varepsilon k}$, using the elementary sine-product integral, and taking $\varepsilon \downarrow 0$ yields (15); the logarithmic diagonal singularity is locally integrable against the smooth integrable factors. Substitute $t = \tan u$, $s = \tan v$, so that $h(t)dt = \cos u\,du$ and

$$\log\frac{t+s}{|t-s|} = \log\frac{\sin(u+v)}{|\sin(u-v)|}.$$

For $0 < u, v < \pi/2$, the classical logarithmic sine Fourier series, interpreted by Abel summation, gives

$$\log\frac{\sin(u+v)}{|\sin(u-v)|} = 2\sum_{n=1}^{\infty}\frac{\sin(2nu)\sin(2nv)}{n}.$$

The integrated coefficients are

$$a_n = \int_0^{\pi/2}\cos u\sin(2nu)\,du = \frac{2n}{4n^2-1}.$$

Abel regularization and the integrability of the logarithmic kernel justify integrating the Fourier series. Therefore

$$\int_0^\infty \frac{H(k)^2}{k}\,dk = \sum_{n=1}^{\infty}\frac{a_n^2}{n} = \sum_{n=1}^{\infty}\frac{4n}{(4n^2-1)^2} = \frac{1}{2}. \tag{16}$$

As $xF(x)^2\,dx = \frac{1}{4}H(k)^2dk/k$ under $k = 2x$, (12) follows. Monotone convergence in (11) gives (13).

For the refined expansion, integration by parts yields

$$F(x) = \frac{1}{2x}\int_0^\infty h(t)\sin(2xt)\,dt.$$

Since $h(0) = 1$, $h''(0) = -3$, and sufficiently many derivatives of h are integrable, repeated integration by parts establishes

$$F(x) = \frac{1}{4x^2} + \frac{3}{16x^4} + O(x^{-6}). \tag{17}$$

Subtract (11) from (13):

$$\frac{1}{2\pi^2} - \Phi(\mu) = \frac{2}{\pi^2}\int_{\mu/2}^\infty (2x-\mu)F(x)^2\,dx. \tag{18}$$

Substituting $F(x)^2 = 1/(16x^4) + 3/(32x^6) + O(x^{-8})$ into (18), with the remainder uniformly integrable over $x \geq \mu/2$, gives

$$\frac{1}{2\pi^2} - \Phi(\mu) = \frac{1}{6\pi^2\mu^2} + \frac{3}{10\pi^2\mu^4} + O(\mu^{-6}),$$

which proves (14). □

5. A uniform modular boundary estimate

Let P_0 be the full-line half-filled reference projector and $B_N = \chi_N P_0 \chi_N$, $W_N = B_N(I - B_N)$. For any self-adjoint finite reference-modular discrepancy E_N, put

$$C_N = \chi_N G_r (I - \chi_N) P_0 \chi_N.$$

The Hilbert–Schmidt Cauchy–Schwarz inequality and the full-line projector identity give

$$2|\operatorname{Tr}(C_N E_N)| \le 2\sqrt{\mathcal{L}_N(r)}\,\sqrt{\operatorname{Tr}(W_N E_N^2)}.$$

Theorem 1 yields the **uniform** inequality

$$\boxed{2|\operatorname{Tr}(C_N E_N)| \le \frac{\sqrt{2}}{\pi}\,\sqrt{\operatorname{Tr}(W_N E_N^2)}.} \tag{19}$$

This coefficient does not depend on r or N. Under the separately established estimate $\operatorname{Tr}(W_N E_N^2) = O(\log N)$, the right-hand side remains $O(\sqrt{\log N})$. It neither establishes $\operatorname{Tr}(S_N E_N) = o(\log N)$ nor proves the leading logarithmic coefficient or finite part of the complete Wilson relative entropy.

6. Mathematical scope

The exact identity (2) uses the stated scalar symbol and even-displacement scalarization. The fixed-r asymptotic (9) and fixed-μ continuum limit (11) are different limiting operations; (9) is not uniform in a mass parameter shrinking with N. The large-μ expansion (14) concerns the continuum function after the finite-interval limit. None of these results addresses the joint sine-kernel/Jacobi moving-index spectrum required for the longitudinal modular estimate.

The separate weighted argument in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* establishes the complete leading law with bounded remainder. The local non-implication statements above describe the scope of the conformal method; they do not leave that leading coefficient unresolved.

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4. K. Justus, *Boundary-Corrected Longitudinal Modular Reduction* (2026). Application to the reference-modular boundary contribution.

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Fixed-Index Wilson Majorant Weights and Longitudinal Edge Contributions

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Abstract

For the half-filled discrete prolate concentration matrix and its commuting finite Jacobi matrix, we prove an explicit cubic asymptotic for the positive Wilson-majorant expectation at every fixed extreme spectral index, with an order-four error bound. The proof combines the finite-Hermite corrected quasimode construction, Fourier concentration, and positivity of the majorant. Exact particle–hole and occupation–majorant relations, together with Slepian's fixed-index concentration asymptotic, determine the contribution of any fixed finite collection of extreme-mode pairs to the longitudinal reference-modular trace. No bound uniform in a growing spectral index is asserted.

1. Operators and conventions

Let $N = 2L$ be even. On $\mathbb{C}^N$, define the symmetric Jacobi matrix

$$(J_N)_{j,j+1} = \frac{(j+1)(N-j-1)}{N}, \qquad (J_N)_{jj} = 0, \quad 0 \le j < N-1.$$

The eigenvalues are simple. Denote its unit real eigenvectors in decreasing eigenvalue order by $v_{N,k}$, $0 \le k < N$. Let P_0 be the infinite-lattice Fourier projection onto $|p| \le \pi/2$, χ_N the projection onto the N-site interval, and $B_N = \chi_N P_0 \chi_N$. In Slepian's notation, $B_N = p(N, 1/4)$ and $J_N = \frac{2}{N}\sigma(N, 1/4)$. His §4.1, equations (129)–(131), pp. 1403–1404, identifies the descending Jacobi eigenvalue index with the descending concentration eigenvalue index. This ordering theorem is a separate classical input; it does not follow merely from commutation.

For $\mu > 0$, set $r_N = \mu/(N+\mu)$. In the interval-preserving scalarization, the full-line positive Wilson majorant has momentum multiplier

$$\widetilde{G}_r(p) = \frac{1}{2}\left(1 - \frac{|\cos p|}{\sqrt{\cos^2 p + r^2 \sin^2 p}}\right), \quad -\pi \le p \le \pi. \tag{1}$$

Write $G_N = \chi_N G_{r_N} \chi_N$ and

$$q_{N,k}(\mu) = \langle v_{N,k}, G_N v_{N,k} \rangle.$$

For every $r > 0$, $0 \le G_r \le I/2$. If $\hat{v}(p) = \sum_{j=0}^{N-1} v_j e^{ijp}$, our Fourier convention is

$$\langle v, G_N v \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} \widetilde{G}_{r_N}(p) |\hat{v}(p)|^2 \, dp. \tag{2}$$

2. Corrected Hermite approximation

Put $h = N^{-1/2}$, $y_j = h(j - (N-1)/2)$, and $H_N = NI/2 - J_N$. The limiting harmonic oscillator is

$$H = -\tfrac{1}{4}\partial_y^2 + 2y^2, \qquad H\psi_k = \varepsilon_k \psi_k, \quad \varepsilon_k = \frac{2k+1}{\sqrt{2}}.$$

For a fixed finite Hermite span, sampling $(\mathcal{S}_h u)_j = \sqrt{h}\, u(y_j)$ gives the lattice expansion

$$H_N \mathcal{S}_h u = \mathcal{S}_h(Hu + h^2 Pu + h^4 Qu) + O_u(h^6),$$

where

$$P = \partial_y(y^2\partial_y) + \tfrac{1}{2} - \tfrac{1}{48}\partial_y^4, \qquad Q = \tfrac{1}{12}\partial_y^2(y^2\partial_y^2) + \tfrac{1}{12}\partial_y^2 - \tfrac{1}{1440}\partial_y^6.$$

The finite Hermite corrections w_k, z_k, orthogonal to ψ_k, solve the successive off-diagonal perturbation equations

$$(H - \varepsilon_k)w_k = -(P - p_k)\psi_k,$$
$$(H - \varepsilon_k)z_k = -((P - p_k)w_k + (Q - \rho_k)\psi_k),$$

with $p_k = \langle \psi_k, P\psi_k\rangle$ and $\rho_k = \langle \psi_k, Q\psi_k\rangle + \langle \psi_k, Pw_k\rangle$. The solvability conditions are precisely the diagonal perturbative equations. After normalization, the sampled vector

$$u_{N,k} = \mathcal{S}_h(\psi_k + h^2 w_k + h^4 z_k)/\|\mathcal{S}_h(\psi_k + h^2 w_k + h^4 z_k)\|$$

obeys a spectral residual $O_k(h^6)$. The fixed-index spectral gap is bounded away from zero for sufficiently small h, by convergence to the simple oscillator eigenvalues. Therefore, with a suitable eigenvector sign,

$$\boxed{\|v_{N,k} - u_{N,k}\|_{\ell^2} = O_k(h^6).} \tag{3}$$

The construction and spectral isolation are established in *Fixed-Index Jacobi Spectral Expansion Through Inverse-Square Order*, §5; equation (3) is the consequence required here. It is not uniform when k grows with N.

3. Cubic expectation theorem

Theorem 1. For each fixed $k \geq 0$ and fixed $\mu > 0$,

$$\boxed{q_{N,k}(\mu) = \frac{\mu^2(2k+1)}{2\sqrt{2}\,N^3} + O_{\mu,k}(N^{-4}).} \tag{4}$$

Proof. Every $u_{N,k}$ is a normalized sample of a fixed finite Hermite combination, with coefficients expanding in even powers of h. Poisson summation and Schwartz decay imply Fourier concentration at $p = O(h)$, with superalgebraically small mass outside each fixed $(-\epsilon, \epsilon)$, $0 < \epsilon < \pi/2$. In particular,

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} p^2 |\hat{u}_{N,k}(p)|^2\, dp = h^2\|\psi_k'\|_2^2 + O_k(h^4),$$
$$\frac{1}{2\pi}\int_{-\pi}^{\pi} p^4 |\hat{u}_{N,k}(p)|^2\, dp = O_k(h^4). \tag{5}$$

On $|p| \leq \epsilon$, with $0 < r \leq 1$, the scalar multiplier has the uniform expansion

$$\widetilde{G}_r(p) = \frac{r^2 p^2}{4} + O_\epsilon(r^2 p^4 + r^4 p^4). \tag{6}$$

As $r_N = \mu h^2 + O_\mu(h^4)$, equations (2), (5), and (6) imply

$$\langle u_{N,k}, G_N u_{N,k}\rangle = \frac{\mu^2 h^6}{4}\|\psi_k'\|_2^2 + O_{\mu,k}(h^8). \tag{7}$$

The oscillator virial identity gives $\|\psi_k'\|_2^2 = \sqrt{2}(2k+1)$.

Positivity provides a sharper eigenvector-transfer estimate than the operator norm alone. From $\|G_N\| \leq 1/2$, equation (3), and (7),

$$\begin{aligned}|\langle v, G_N v\rangle - \langle u, G_N u\rangle| &\leq 2\|G_N^{1/2}u\|\,\|G_N^{1/2}(v-u)\| + \|G_N^{1/2}(v-u)\|^2 \\ &= O_{\mu,k}(h^9) + O_k(h^{12}).\end{aligned}$$

Together with (7) this gives an $O(h^8) = O(N^{-4})$ remainder and proves (4). □

The estimate depends on the third-order Hermite quasimode and remains a fixed-index theorem. A standard unweighted $\|v - u\|$ estimate alone would not provide the same conclusion without positivity.

4. Exact occupation bounds and edge-pair contribution

Let $S = G(I - 2P_0)$, $S_N = \chi_N S \chi_N$. For a unit interval-supported vector v, write $b = \langle v, P_0 v\rangle$, $q = \langle v, Gv\rangle$, $s = \langle v, Sv\rangle$. Since $[G, P_0] = 0$ and $0 \le 2G \le I$,

$$\boxed{0 \le q + s \le 1 - b, \qquad 0 \le q - s \le b.} \tag{8}$$

Indeed, $q + s = 2\langle (I - P_0)v, G(I - P_0)v\rangle$ and $q - s = 2\langle P_0 v, GP_0 v\rangle$.

Define $D = \operatorname{diag}((-1)^j)_{j=0}^{N-1}$. Directly from the half-filled kernel and the even-displacement majorant,

$$DJ_N D = -J_N, \quad DB_N D = I - B_N, \quad DG_N D = G_N, \quad DS_N D = -S_N. \tag{9}$$

If η is the odd reference scaling function, the self-adjoint discrepancy

$$E_N = \log(I - B_N) - \log B_N + N\eta(J_N/N)$$

also satisfies $DE_N D = -E_N$. Thus the bottom paired Jacobi vector has $b \mapsto 1 - b$, $q \mapsto q$, $s \mapsto -s$ and $e \mapsto -e$, with no ambiguity in the pair signs.

The classical Slepian fixed-index concentration asymptotic, with the discrete-prolate index correspondence, gives

$$1 - b_{N,k} = O_k(N^{k+1/2} e^{-2N \operatorname{arsinh} 1}) \quad \text{for fixed } k.$$

For each fixed k, the paired modular discrepancy has the finite limit

$$\lim_{N\to\infty} e_{N,k} = c_k = \tfrac{1}{2}\log\pi - \log(k!) - k\log 2 + (k + \tfrac{1}{2})[\log(2k+1) - 1]. \tag{10}$$

Theorem 2. For every fixed positive integer K, the top and bottom K pairs contribute to the original longitudinal trace by

$$\boxed{\sum_{k=0}^{K-1} (s_{N,k} e_{N,k} + s_{N,N-1-k} e_{N,N-1-k}) = -\frac{\mu^2}{\sqrt{2}\, N^3} \sum_{k=0}^{K-1} (2k+1) c_k + o_{\mu,K}(N^{-3}).} \tag{11}$$

Proof. From (8), $s_{N,k} = -q_{N,k} + O(1 - b_{N,k})$ at each fixed top index. The complementary occupations decay exponentially in N, so Theorem 1 and (10) imply $s_{N,k} e_{N,k} = -\mu^2 (2k + 1) c_k / (2\sqrt{2} N^3) + o(N^{-3})$. Equation (9) gives an identical contribution from the paired bottom index. Summing over the fixed set proves (11). □

5. Scope of the result

The fixed-index error constants may depend on k and μ, and no estimate is asserted for $K = K(N) \to \infty$. In particular, neither the fixed-index results nor the conformal boundary-energy bound establishes by itself the longitudinal estimate

$$\operatorname{Tr}(S_N E_N) = o_\mu(\log N).$$

The separate argument in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* establishes this longitudinal estimate with the stronger bound $O_\mu(1)$ and yields the leading entropy law. For the specified fixed-μ Wilson comparison, the finite part is Theorem 1 of *Finite Part of Wilson–Dirac Relative Entropy*. Corollary 4 of *Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals* strengthens the longitudinal estimate to $O_\mu(N^{-1})$. Neither conclusion supplies uniform error bounds for the present fixed-index formulas with $K = K(N) \to \infty$. Numerical tests are supplementary evidence, not interval-certified enclosures or proof of moving-index uniformity.

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A Positive Reference-Modular Trace for Wilson–Dirac Intervals

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Abstract

We formulate the longitudinal reference-modular contribution to the quasifree relative entropy of a Wilson–Dirac interval in terms of a positive finite-matrix trace. A particle–hole pairing argument bounds the difference between the signed longitudinal generator and a positive reference logarithm. Together with an optimal scalar cubic envelope, a Wilson-majorant spectral-tail estimate, and the exact longitudinal Jacobi first moment, this yields a positive-trace reduction with an explicit one-sided, finite-dimensional error bound. The leading logarithmic asymptotic of that positive trace is **not** evaluated here. The full fixed-positive-scaled-mass entropy expansion is proved separately in Theorem 1 of *Finite Part of Wilson–Dirac Relative Entropy*, with a finite boundary trace path and remainder $O_\mu(1/\log L)$. This positive-trace reduction alone is not that proof.

1. Finite reference and operator hypotheses

For an even number of scalarized sites, $N = 2L$, consider the half-filled sine-kernel compression B_N, the commuting tridiagonal Jacobi matrix J_N, and

$$W_N = B_N(I - B_N), \qquad K_N = \log(I - B_N) - \log B_N, \qquad E_N = K_N + N\eta(J_N/N).$$

The infinite-line occupied-band projection is P_0, compressed by the interval projection χ_N to B_N. The full-line Wilson majorant G and longitudinal perturbation S obey

$$G \geq 0, \qquad 0 \leq 2G \leq I, \qquad [G, P_0] = 0, \qquad S = G(I - 2P_0).$$

Write $G_N = \chi_N G \chi_N$ and $S_N = \chi_N S \chi_N$. Finite compression preserves operator inequalities but **not** products. Let $D = \mathrm{diag}((-1)^j)$; the half-filled symmetry gives

$$DB_N D = I - B_N, \quad DJ_N D = -J_N, \quad DG_N D = G_N, \quad DS_N D = -S_N.$$

The reference has $0 < B_N < I$, and it has no occupation eigenvalue equal to $1/2$ when N is even: after reordering by site parity, $B_N - I/2$ is off-diagonal with an invertible Cauchy block. Thus its upper and lower occupation sectors are exactly paired. Define $H_N = \mathrm{sgn}(2B_N - I)$.

We use the following separately established finite-interval estimates, with $r = \mu/(N+\mu)$, $\mu > 0$ fixed:

$$\begin{aligned} M_N &:= \operatorname{Tr} G_N = N\left[\frac{1}{2} - \frac{\arccos r}{\pi\sqrt{1-r^2}}\right] \leq \frac{\mu}{\pi}, \\ \nu_N((t,\infty)) &:= \sum_{|\lambda_j|>t} \langle v_j, G_N v_j\rangle \leq \frac{a_\mu}{t} + \frac{b_\mu}{t^2}, \\ a_\mu &= \sqrt{\frac{2}{15}}\frac{\mu^2}{\pi}, \qquad b_\mu = \frac{\mu}{3\pi}, \end{aligned} \tag{1}$$

where $J_N v_j = \lambda_j v_j$. The reference-crossing trace-class estimate is

$$\operatorname{Tr}\sqrt{W_N} \le \frac{2}{\pi} I_N, \quad I_N = \int_0^\infty e^{-t} \frac{\sqrt{1-e^{-2Nt}}}{1-e^{-2t}}\, dt = \frac{1}{2}\int_0^1 \frac{\sqrt{1-u^N}}{\sqrt{u}(1-u)}\, du = O(\log N). \tag{2}$$

The complete relative-entropy reduction is a separate mathematical input:

$$\mathcal{D}_L(\mu) = \frac{\mu^2}{6}\log L + \operatorname{Tr}(S_N E_N) + O_\mu(\sqrt{\log L}). \tag{3}$$

2. Positive reference-logarithm comparison

Lemma 1. For any self-adjoint F_N satisfying $[F_N, B_N] = 0$ and $DF_N D = -F_N$,

$$|\operatorname{Tr}(S_N F_N) + \operatorname{Tr}(G_N H_N F_N)| \le 2\sqrt{M_N \operatorname{Tr}(W_N F_N^2)}. \tag{4}$$

Proof. Choose a paired common eigenbasis of B_N, F_N. For the eigenvector in the upper occupation sector write $b > 1/2$, f for the F_N-eigenvalue, $q = \langle v, G_N v\rangle$, $s = \langle v, S_N v\rangle$, and $\theta = q + s$. The partner Dv has eigenvalues $1-b, -f$ and expectations $q, -s$. On the full line, $G + S = 2G(I - P_0)$ is positive and dominated both by $I - P_0$ and $2G$. Compression of these *inequalities* gives

$$0 \le \theta \le \min(1-b, 2q).$$

Hence, for $w = b(1-b)$, $\theta^2/w \le \theta/b \le 2\theta$. Summing over upper occupations and using pair symmetry,

$$\sum_+ \frac{\theta_j^2}{w_j} \le 2M_N, \quad \sum_+ w_j f_j^2 = \tfrac{1}{2}\operatorname{Tr}(W_N F_N^2), \quad \operatorname{Tr}(S_N F_N) + \operatorname{Tr}(G_N H_N F_N) = 2\sum_+ \theta_j f_j.$$

Cauchy–Schwarz proves (4). In particular, no commutation of G_N with B_N has been assumed. □

Take $F_N = K_N$. Since $H_N K_N = -|K_N|$ and the upper occupation eigenvalue of K_N is negative, its paired trace has a fixed sign:

$$\boxed{\begin{aligned} &0 \le P_N - \operatorname{Tr}(S_N K_N) \le U_N := 2\sqrt{M_N T_{K,N}}, \\ &P_N := \operatorname{Tr}(G_N |K_N|) \ge 0, \quad T_{K,N} := \operatorname{Tr}(W_N K_N^2). \end{aligned}} \tag{5}$$

The sharp scalar occupation-logarithm theorem, proved in the companion study *Finite-Interval Occupation Logarithm and Reference-Crossing Bounds*, gives $b(1-b)\log^2((1-b)/b) \le C_{\log}\sqrt{b(1-b)}$, with $C_{\log} = 2x_*^2/\cosh x_*$, $x_* \tanh x_* = 2$. The same study proves $I_N \le F_N := \operatorname{arsinh}\sqrt{N-1} + \sqrt{N}\arcsin(N^{-1/2})$. Hence

$$\boxed{U_N \le 2\sqrt{\frac{2C_{\log} M_N F_N}{\pi}} = O_\mu(\sqrt{\log N}).} \tag{6}$$

3. Uniformly bounded nonlinear Jacobi contribution

The reference scaling function has the positive-coefficient integral series

$$\eta(t) = 4t\int_0^1 \frac{\sqrt{1-u^2}}{\sqrt{1-4t^2u^4}}\, du = \pi t + \sum_{j\ge 1} c_j t^{2j+1}, \qquad c_j \ge 0, \quad |t| \le \frac{1}{2}.$$

Since $\eta(1/2) = 2\,\mathrm{arsinh}(1)$, the optimal global cubic envelope on this scalar interval is

$$\boxed{|\eta(t) - \pi t| \le C_\eta |t|^3, \quad C_\eta = 16\,\mathrm{arsinh}(1) - 4\pi.} \tag{7}$$

From the tail bound (1) and layer cake,

$$\frac{1}{N^2}\sum_j q_j|\lambda_j|^3 = \frac{3}{N^2}\int_0^{N/2} t^2 \nu_N((t,\infty))dt \le \frac{3a_\mu}{8} + \frac{3b_\mu}{2N}.$$

The operator order $-G_N \le S_N \le G_N$ then implies

$$\boxed{|\mathrm{Tr}\,[S_N(N\eta(J_N/N) - \pi J_N)]| \le V_{N,\mu} := C_\eta\left(\frac{3a_\mu}{8} + \frac{3b_\mu}{2N}\right) = O_\mu(1).} \tag{8}$$

This bounds a **majorant-weighted trace**, not the operator-norm difference between the two generators.

4. Finite-dimensional enclosure and asymptotic reduction

Theorem 2 (positive trace and directional error). For every even $N = 2L$ and fixed positive μ, subject to the hypotheses and estimates in §1,

$$\boxed{P_N + \pi\,\mathrm{Tr}(S_N J_N) - U_N - V_{N,\mu} \le \mathrm{Tr}(S_N E_N) \le P_N + \pi\,\mathrm{Tr}(S_N J_N) + V_{N,\mu}.} \tag{9}$$

The fully explicit upper envelope (6) may replace U_N if $T_{K,N}$ is not evaluated. The right-hand deviation from $P_N + \pi\,\mathrm{Tr}(S_N J_N)$ is bounded uniformly in N; only the lower deviation is presently controlled at $O_\mu(\sqrt{\log N})$.

Proof. Decompose $E_N = K_N + \pi J_N + [N\eta(J_N/N) - \pi J_N]$. Apply the signed inequality (5) to the first term and (8) to the final term. □

The companion exact Jacobi moment is

$$\mathrm{Tr}(S_N J_N) = \frac{N^2-1}{3\pi}\left[\frac{\mathbf{E}(1-r^2) - r^2\mathbf{K}(1-r^2)}{1-r^2} - 1\right],$$

with the elliptic **parameter** $1 - r^2$. NIST DLMF §19.12 gives its fixed-μ expansion

$$\pi\,\mathrm{Tr}(S_N J_N) = -\frac{\mu^2}{6}\log N + \frac{\mu^2}{4} - \frac{\mu^2}{6}\log\frac{4}{\mu} + O_\mu\left(\frac{\log N}{N}\right).$$

Combining with the separate entropy reduction (3), and $\log N = \log L + \log 2$, proves the positive-trace formulation

$$\boxed{\mathcal{D}_L(\mu) = P_N + \frac{\mu^2}{4} - \frac{\mu^2}{6}\log\frac{8}{\mu} + O_\mu(\sqrt{\log N}).} \tag{10}$$

Equation (10) **does not identify the full entropy finite part**. The error is not known to converge. The complete leading coefficient $\mu^2/6$ is equivalent, through (10), to the positive-trace condition below. It follows from the leading entropy theorem in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*:

$$\boxed{\mathrm{Tr}(G_N|K_N|) = \frac{\mu^2}{6}\log N + o_\mu(\log N).} \tag{11}$$

5. Spectral-measure formulation

Put $A_N = |2B_N - I|$, whose spectrum lies in $[0, 1)$. Spectral calculus yields

$$P_N = 2\int_0^1 \mathrm{Tr}[G_N A_N (I - t^2 A_N^2)^{-1}]\, dt = \int_0^\infty \mathrm{Tr}[G_N \mathbf{1}_{\{|K_N|>s\}}] ds. \tag{12}$$

These positive operator-integral formulations require no Jacobi eigenvector ordering. They formulate the trace in terms of the weighted reference spectral measure. Its logarithmic growth is determined by the separate weighted asymptotic argument cited above; positivity alone does not determine the coefficient.

6. Limitations and mathematical references

The statements refer to infinite-chain band projections compressed to a finite interval and the fixed scaled-mass regime $m = \mu/L$. They do not apply automatically to finite open-boundary ground states or establish estimates uniform in a moving spectral index. The leading entropy theorem uses the separate compact-interior and weighted modular bounds. A finite part and an inverse-logarithmic convergence rate are not established by the present reduction.

The finite-interval majorant, reference crossing and entropy results invoked in §1 are mathematical inputs of the companion studies *Exact Spectral Majorants for a Wilson–Dirac Interval*, *Logarithmic Weighted Modular Control for the Half-Filled Sine Kernel*, *Exact Longitudinal Jacobi Moment and Wilson Majorant Mass*, and *Finite-Interval Wilson–Dirac Relative Entropy*, together with the sharp bound in *Finite-Interval Occupation Logarithm and Reference-Crossing Bounds*. The exact elliptic asymptotic is consistent with NIST Digital Library of Mathematical Functions, §19.12 (https://dlmf.nist.gov/19.12). Classical discrete-prolate context includes Slepian's 1978 discrete case and Grünbaum's 1981 commuting-tridiagonal results. Neither classical reference is asserted to establish this Wilson-specific theorem.

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Positive Reference-Modular Localization and Bounds for the Wilson Majorant Jacobi Moment

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Abstract

For the half-filled finite sine-kernel reference and a Wilson positive majorant, we prove a parameterized spectral-wing bound for the positive reference-modular trace and derive complementary upper and lower logarithmic bounds for the positive majorant-weighted Jacobi moment. The arguments use finite-dimensional order inequalities, the established Wilson majorant tail, the exact signed Jacobi first moment, and a published discrete-prolate eigenvalue bound. No uniform finite-size approximation of the reference modular generator by the Jacobi generator is assumed. The full fixed-positive-scaled-mass entropy expansion is proved separately in Theorem 1 of *Finite Part of Wilson–Dirac Relative Entropy*, with remainder $O_\mu(1/\log L)$. Theorem 3 of *Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals* proves a finite-mode bound $\|E_N\| \le 7$; it does not assert a uniform moving-edge expansion.

1. Operators and mathematical inputs

Let $N = 2L$ be even and $\mu > 0$ fixed. The reference covariance and commuting Jacobi matrix are

$$B_N(i,j) = \begin{cases} \frac{1}{2} & i = j, \\ \frac{\sin[\pi(i-j)/2]}{\pi(i-j)} & i \neq j, \end{cases} \qquad (J_N)_{j-1,j} = \frac{j(N-j)}{N},$$

with all other Jacobi entries zero. The reference modular operator and the positive Wilson majorant are

$$K_N = \log(I - B_N) - \log B_N, \qquad G_N \succeq 0, \qquad M_N = \operatorname{Tr} G_N \le \frac{\mu}{\pi}.$$

Write $S_N = S_N^*$ for the longitudinal Wilson covariance perturbation and assume the exact operator inequalities $-G_N \preceq S_N \preceq G_N$. The following established majorant tail is used as a mathematical input, for every $t > 0$:

$$\nu_N((t,\infty)) := \operatorname{Tr}\big(G_N \mathbf{1}_{\{|J_N| > t\}}\big) \le \frac{a_\mu}{t} + \frac{b_\mu}{t^2}, \quad a_\mu = \sqrt{\frac{2}{15}}\frac{\mu^2}{\pi}, \quad b_\mu = \frac{\mu}{3\pi}. \tag{1}$$

The exact longitudinal first moment has the asymptotic

$$\operatorname{Tr}(S_N J_N) = -\frac{\mu^2}{6\pi}\log N + \frac{\mu^2}{4\pi} - \frac{\mu^2}{6\pi}\log\frac{4}{\mu} + O_\mu\!\left(\frac{\log N}{N}\right). \tag{2}$$

The published discrete-prolate lower eigenvalue estimate, combined with half-filled particle–hole symmetry, gives

$$\|K_N\| \le 2N \operatorname{arsinh}(1) + 3\log N + C \tag{3}$$

for an absolute C and even N. The full Wilson entropy is already reduced, under its separately established hypotheses, to

$$\mathcal{D}_L(\mu) = P_N + C_\mu + O_\mu(\sqrt{\log N}), \quad P_N = \mathrm{Tr}(G_N|K_N|), \quad C_\mu = \frac{\mu^2}{4} - \frac{\mu^2}{6}\log\frac{8}{\mu}. \tag{4}$$

The explicit C_μ in (4) is **not** a certified entropy finite part: the displayed remainder has not been shown to converge. Equations (1)–(4) are inputs from the cited studies, not independently re-proved in this manuscript.

2. A spectral-wing trace inequality

Lemma 1 (order domination). Let $G \succeq 0$, $-G \preceq S = S^* \preceq G$, and $F = F^*$ be finite matrices. Then

$$\boxed{\left|\mathrm{Tr}(SF)\right| \le \mathrm{Tr}(G|F|).} \tag{5}$$

Proof. Decompose $F = F_+ - F_-$ spectrally, with $F_\pm \succeq 0$. Trace monotonicity against each positive part gives $\left|\mathrm{Tr}(SF_\pm)\right| \le \mathrm{Tr}(GF_\pm)$. The triangle inequality yields (5). No commutativity between G and F is required. □

Let $\Pi_{>R} = \mathbf{1}_{\{|J_N|>R\}}$, $\Pi_{\le R} = I - \Pi_{>R}$, and $P_N^{\le R} = \mathrm{Tr}(G_N|K_N|\Pi_{\le R})$. Because $[B_N, J_N] = 0$, the operators $|K_N|$ and $\Pi_{>R}$ commute and their product is positive.

Theorem 2 (quantitative positive-trace localization). Under (1) and (3), for all $R > 0$,

$$\boxed{0 \le P_N - P_N^{\le R} \le \left(2N\,\mathrm{arsinh}(1) + 3\log N + C\right)\left(\frac{a_\mu}{R} + \frac{b_\mu}{R^2}\right).} \tag{6}$$

Proof. Positivity follows from $\mathrm{Tr}(G_N X) \ge 0$ whenever $G_N, X \succeq 0$, taking $X = |K_N|\Pi_{>R}$. Since $0 \preceq X \preceq \|K_N\|\Pi_{>R}$, trace monotonicity, (1), and (3) imply (6). □

Corollary 3 (adjustable sublogarithmic window). Let $h_N \to \infty$ with $h_N = o(\log N)$, and set $R_N = N/h_N$. Then

$$\boxed{P_N - P_N^{\le R_N} = O_\mu(h_N) = o_\mu(\log N),} \tag{7}$$

and (4) becomes

$$\boxed{\mathcal{D}_L(\mu) = P_N^{\le N/h_N} + C_\mu + O_\mu(\sqrt{\log N} + h_N).} \tag{8}$$

Proof. Substitute $R_N = N/h_N$ in (6). The leading term is $2a_\mu\,\mathrm{arsinh}(1)h_N$, with lower-order contributions $O_\mu(h_N \log N/N + h_N^2/N)$, hence (7). Equation (8) follows from (4). □

For example, $h_N = \log N/\log\log N$ produces the narrower region $|J_N| \le N\log\log N/\log N$ while the discarded wing remains $o(\log N)$. This choice **worsens the stated remainder** relative to $h_N = \sqrt{\log N}$; it only tightens the window relevant for a leading-coefficient proof. In general the coarse bound requires $R_N \gg N/\log N$. This is a sufficient scale for this method, not a necessary physical threshold.

The desired complete leading logarithmic coefficient is equivalent, under (4), to

$$\boxed{P_N^{\le N/h_N} = \frac{\mu^2}{6}\log N + o_\mu(\log N).} \tag{9}$$

Equation (9) follows by combining the separate leading entropy law and positive-trace reduction with the discarded-wing estimate above. The finite-dimensional wing bound alone does not determine its coefficient.

3. Positive majorant-weighted Jacobi moment

Define $Q_N(\mu) = \operatorname{Tr}(G_N|J_N|)$. Since J_N is a self-adjoint Jacobi matrix and $\|J_N\| \le N/2$, the layer-cake representation gives

$$Q_N(\mu) = \int_0^{N/2} \nu_N((t,\infty))\, dt. \tag{10}$$

Theorem 4 (logarithmic coefficient bracket). For every fixed $\mu > 0$,

$$\boxed{\frac{\mu^2}{6\pi} \le \liminf_{N\to\infty,\ N\,\text{even}} \frac{Q_N(\mu)}{\log N} \le \limsup_{N\to\infty,\ N\,\text{even}} \frac{Q_N(\mu)}{\log N} \le \sqrt{\frac{2}{15}}\frac{\mu^2}{\pi}.} \tag{11}$$

Proof. Lemma 1 with $F = J_N$ gives $Q_N \ge |\operatorname{Tr}(S_N J_N)|$. The exact asymptotic (2) then proves the lower limit. For the upper limit, split (10) at $t = 1$. For $N \ge 2$, the interval $(0,1)$ contributes at most M_N, while (1) gives

$$Q_N \le M_N + a_\mu \log(N/2) + b_\mu(1 - 2/N).$$

Divide by $\log N$ and let $N \to \infty$, using $M_N \le \mu/\pi$. □

The lower bound already has the coefficient $\mu^2/(6\pi)$ anticipated for Q_N, but the majorant-tail estimate alone does not establish a matching upper bound. A numerical observation that $Q_N + \operatorname{Tr}(S_N J_N)$ remains small over tested finite sizes is not a proof of boundedness or decay.

4. A sufficient two-part proof obligation

The positive-trace problem can be decomposed *algebraically* as

$$P_N = \pi Q_N + \underbrace{\operatorname{Tr}\big[G_N(|K_N| - \pi|J_N|)\big]}_{\Delta_N^{\mathrm{abs}}}. \tag{12}$$

Therefore the pair of estimates

$$Q_N = \frac{\mu^2}{6\pi}\log N + o_\mu(\log N), \qquad \Delta_N^{\mathrm{abs}} = o_\mu(\log N) \tag{13}$$

would imply (9), and hence the full leading relative-entropy coefficient. These conditions are sufficient, **not individually necessary**, because cancellations between two terms cannot be ruled out. Neither estimate in (13) is proved here.

The positive operator structure and signed exact Jacobi moment thus give a precise framework for separating majorant spectral-weight growth from the modular comparison. The separate weighted theorem in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* gives $P_N = \pi Q_N + O_\mu(1)$; combined with the proved entropy law and the positive-trace reduction, this also yields $Q_N = (\mu^2/(6\pi))\log N + O_\mu(\sqrt{\log N})$.

5. Scientific limitations and references

All operator statements concern the infinite-chain occupied-band projectors compressed to the specified finite interval. The positive-majorant tail, entropy reduction, and elliptic first moment are mathematical inputs from the named companion papers. Numerical experiments on finite matrices are corroboration only. Fixed-index discrete-prolate asymptotics must not be substituted for estimates uniform over moving spectral indices. The finite-part convergence and any uniform moving-edge expansion are separate from the established leading entropy coefficient.

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From Prephysical Hypotheses to Physical Models

A Uniform Sobolev Bound for the Continuum Cross-Block Interpolants

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Abstract

The entropy-difference estimate (B9) of *Cross-Boundary Trace-Class Estimates for Wilson–Dirac Quasifree States* uses a constant M_s that bounds the continuum cross-block interpolants on a fractional Sobolev space. This note defines the relevant space, proves $M_s \le \frac{1}{2} + \max\{c_s, c'_s\}$ for $0 \le s < 1/2$, where c_s and c'_s are the pointwise-multiplier norms of the indicators of the unit interval and of its complement on $H^s(\mathbb{R})$, and records that the bound is independent of the interpolation parameter, the scaled mass and the lattice size. The only external input is the classical multiplier theorem for indicators of half-lines.

1 Setting

Notation follows [1]. Let $H^s = H^s(\mathbb{R};\mathbb{C}^2)$ with $\|f\|_s^2 = \int_{\mathbb{R}} (1+u^2)^s |\hat{f}(u)|^2\, du$, and $\chi = \mathbf{1}_{(0,1)}$. For $b \ge 0$ let P_b^c be the Fourier multiplier with symbol

$$p_b(u) = \tfrac{1}{2}\Big[I - \frac{u\sigma_x + b\sigma_z}{\sqrt{u^2+b^2}}\Big],$$

an orthogonal projection on $\mathbb{C}^2$ for every $(u,b) \ne (0,0)$. Put $X_b = \chi P_b^c (1-\chi)$, $Z_\mu = X_\mu - X_0$ and

$$\mathbb{H}(X) = \begin{pmatrix} 0 & X \\ X^* & 0 \end{pmatrix}, \qquad \mathbb{D}_\mu = \mathbb{H}(Z_\mu), \qquad A_\theta = \mathbb{H}(X_0) + \theta\, \mathbb{D}_\mu, \qquad Y_\theta = A_\theta + \tfrac{1}{2} I, \qquad 0 \le \theta \le 1.$$

The block form refers to $L^2(\mathbb{R};\mathbb{C}^2) = L^2((0,1);\mathbb{C}^2) \oplus L^2(\mathbb{R}\setminus(0,1);\mathbb{C}^2)$, $f = f_{\text{in}} + f_{\text{out}}$ with $f_{\text{in}} = \chi f$. Since $A_\theta = (1-\theta)\mathbb{H}(X_0) + \theta\, \mathbb{H}(X_\mu)$,

$$\mathbb{H}(X_b) f = \chi P_b^c f_{\text{out}} \;+\; (1-\chi) P_b^c f_{\text{in}}. \tag{1}$$

Definition 1 (Block Sobolev space)**.** For $0 \le s < 1/2$ let $\mathcal{H}^s$ be the space of $f \in L^2(\mathbb{R};\mathbb{C}^2)$ whose parts $f_{\text{in}}, f_{\text{out}}$, extended by zero, belong to H^s, with $\|f\|_{\mathcal{H}^s}^2 = \|f_{\text{in}}\|_s^2 + \|f_{\text{out}}\|_s^2$.

The classical theorem of Shamir and Strichartz [2, 3] states that multiplication by the indicator of a half-line is bounded on $H^s(\mathbb{R})$ exactly for $-1/2 < s < 1/2$. Let

κ_s be the norm of multiplication by $\mathbf{1}_{(0,\infty)}$ on $H^s(\mathbb{R})$, $0 \le s < 1/2$. The H^s norm is invariant under translation and reflection, and $\chi = \mathbf{1}_{(0,\infty)} - \mathbf{1}_{[1,\infty)}$. Hence the multiplier norms

$$c_s = \|\chi\|_{H^s \to H^s} \le 2\kappa_s, \qquad c'_s = \|1-\chi\|_{H^s \to H^s} \le 1 + 2\kappa_s$$

are finite, and both are at least one because each multiplier fixes smooth functions supported inside $(0,1)$, respectively outside $[0,1]$. Consequently $\mathcal{H}^s = H^s$ as sets, with $(c_s^2 + c_s'^2)^{-1/2}\|f\|_{\mathcal{H}^s} \le \|f\|_s \le \sqrt{2}\,\|f\|_{\mathcal{H}^s}$.

2 The bound

Lemma 2 (Projection multipliers)**.** *For every $b \ge 0$ and $s \in \mathbb{R}$, $\|P_b^c f\|_s \le \|f\|_s$.*

Proof. $\widehat{P_b^c f}(u) = p_b(u)\hat{f}(u)$ and $|p_b(u)v| \le |v|$ for almost every u; the value at $(u,b) = (0,0)$ is irrelevant. Multiply by $(1+u^2)^s$ and integrate. □

Theorem 3 (Uniform bound for the interpolants)**.** *Let $0 \le s < 1/2$. For all $b \ge 0$,*

$$\|\mathbb{H}(X_b)\|_{\mathcal{H}^s \to \mathcal{H}^s} \le \max\{c_s, c'_s\}.$$

Consequently, for all $\theta \in [0,1]$ and all $\mu \ge 0$,

$$\|Y_\theta\|_{\mathcal{H}^s \to \mathcal{H}^s} \le M_s := \tfrac{1}{2} + \max\{c_s, c'_s\} \le \tfrac{3}{2} + 2\kappa_s,$$

and M_s depends only on s.

Proof. By (1) the inner part of $\mathbb{H}(X_b)f$ is $\chi P_b^c f_{\text{out}}$ and its outer part is $(1-\chi)P_b^c f_{\text{in}}$. Lemma 2 and the definitions of c_s, c'_s give $\|\chi P_b^c f_{\text{out}}\|_s \le c_s\|f_{\text{out}}\|_s$ and $\|(1-\chi)P_b^c f_{\text{in}}\|_s \le c'_s\|f_{\text{in}}\|_s$. Therefore $\|\mathbb{H}(X_b)f\|_{\mathcal{H}^s}^2 \le c_s^2\|f_{\text{out}}\|_s^2 + c_s'^2\|f_{\text{in}}\|_s^2 \le \max\{c_s, c'_s\}^2\|f\|_{\mathcal{H}^s}^2$. The second claim follows from $Y_\theta = \frac{1}{2}I + (1-\theta)\mathbb{H}(X_0) + \theta\,\mathbb{H}(X_\mu)$ and the triangle inequality. □

Corollary 4 (Use in the entropy-difference estimate)**.** *Let $0 < s < 1/2$, let $Y_{L,\theta}$ be the lattice interpolant of [1], and assume its estimate (B8), $\|\widetilde{P}_{L,\mu} - P_\mu^c\|_{H^s \to L^2} \le C_s L^{-s}$ uniformly in $\mu \ge 0$. Then for integers $k \ge 1$,*

$$\left\|(Y_{L,\theta}^k - Y_\theta^k)\mathbb{D}_\mu\right\|_1 \le \sqrt{2}\,C_s\,k\,M_s^{k-1}L^{-s}\,\|\mathbb{D}_\mu\|_{\mathcal{S}_1(L^2,\mathcal{H}^s)}.$$

If a polynomial $\sum_{k\le n} a_k Y^k$ has $\sum_k |a_k| \le C7^n$, the corresponding difference of traces against $\mathbb{D}_\mu$ is at most $\sqrt{2}\,C\,C_s\,n\,(7M_s)^n L^{-s}\|\mathbb{D}_\mu\|_{\mathcal{S}_1(L^2,\mathcal{H}^s)}$. With $s = 1/4$ and $n = \lfloor \log L/(8\log(7M_{1/4}))\rfloor$ one has $(7M_{1/4})^n \le L^{1/8}$, so this term is $O(n\,L^{-1/8})$ times the nuclear norm, uniformly in θ and μ.

Proof. Write $Y_L^k - Y^k = \sum_{j=0}^{k-1} Y_L^j (Y_L - Y) Y^{k-1-j}$. The lattice interpolants are L^2 contractions [1]. By (1) and its lattice analogue, $(Y_L - Y)f$ is a convex combination of terms $\chi(\widetilde{P}-P^c)f_{\text{out}} + (1-\chi)(\widetilde{P}-P^c)f_{\text{in}}$, whose L^2 norm is at most $C_sL^{-s}(\|f_{\text{out}}\|_s + \|f_{\text{in}}\|_s) \le \sqrt{2}\,C_sL^{-s}\|f\|_{\mathcal{H}^s}$. Theorem 3 bounds Y^{k-1-j} on $\mathcal{H}^s$ by $M_s^{k-1-j} \le M_s^{k-1}$, since $M_s \ge 3/2$. The ideal property of the nuclear norm gives the first display; summing over $k \le n$ gives the second. The choice of n gives $n\log(7M_{1/4}) \le \frac{1}{8}\log L$. □

Scope. The theorem supplies the constant written M_s in the display preceding (B9) of [1] and shows that it does not depend on θ, μ or L. The corollary uses (B8), the contraction property of the lattice interpolants and the polynomial coefficient bound as stated there; it does not re-prove them, the nuclear estimate (B7), the trace-norm comparison (B6) or the Chebyshev approximation rate. The constants c_s, c_s' are not evaluated numerically here. A finite-grid computation at $s = 1/4$ (accompanying program `check_Ms_bound.py`) is consistent with the inequality; it is a sanity check and not part of the proof.

Research and computational disclosure

This is an AI-assisted mathematical research manuscript prepared within the research program *From Prephysical Hypotheses to Physical Models*. Research direction and scholarly responsibility: Kirtus Justus. The argument was drafted by Claude (Anthropic) at the author's direction and is presented for examination; it has not been independently refereed. Assistance does not imply provider endorsement.

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